Supervised by:
Dr. Mohammed Sharkawy

Hematology
2026

Out of the League
Family Medicine

Prepared by: Dr. Haya Alturkait

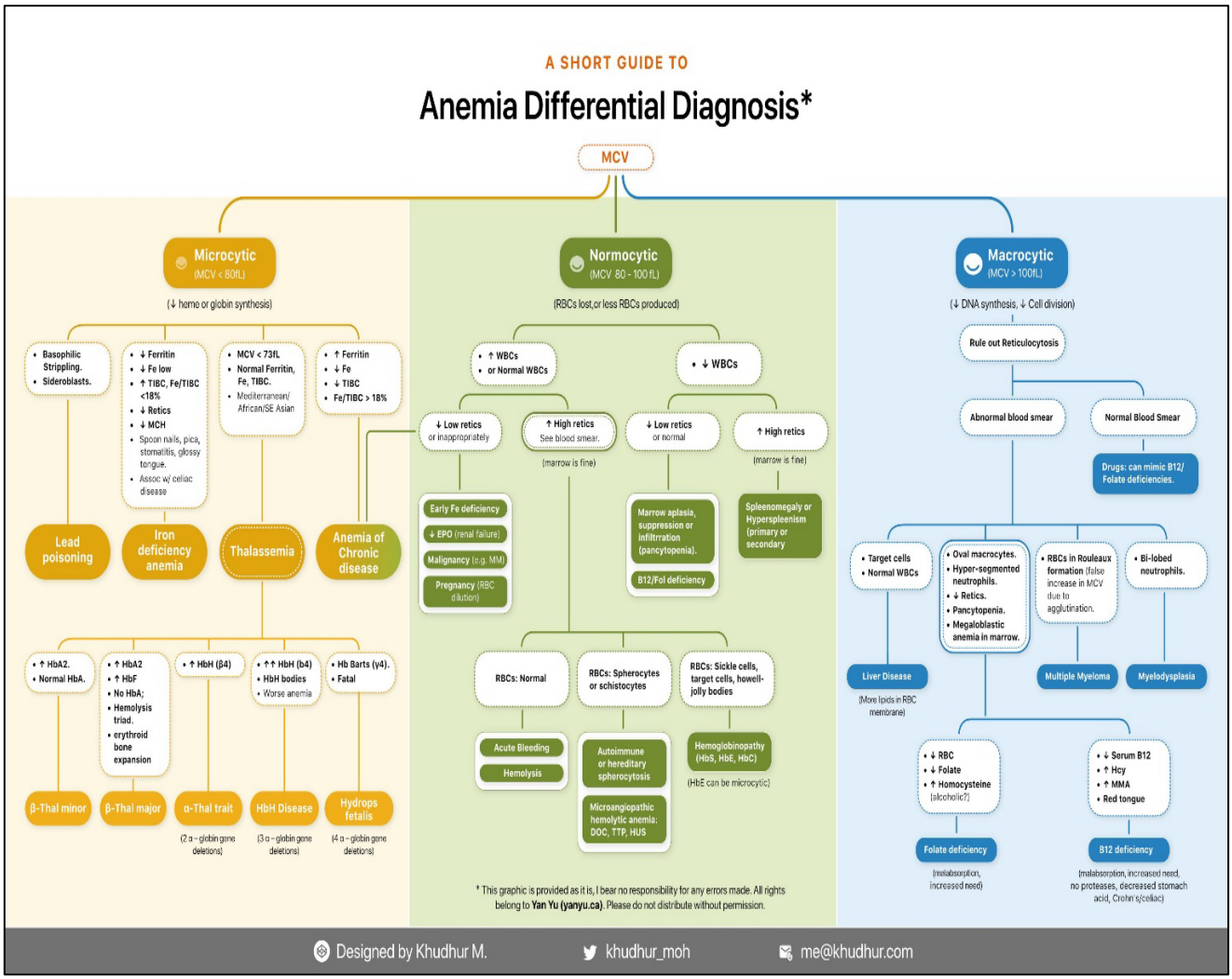
3rd Edition

Table of Contents

TOPIC: Anemia ***4***
TOPIC: Macrocytic anemia ***5***
TOPIC: Anemia - B12 and folate deficiency ***6***
TOPIC: Microcytic anaemia ***16***
TOPIC: Anaemia – iron deficiency: ***17***
TOPIC: Normocytic anaemia ***28***
TOPIC: Lead poisoning ***29***
TOPIC: Autoimmune haemolytic anaemia ***31***
TOPIC: Fanconi anaemia ***32***
TOPIC: G6PD deficiency ***33***
TOPIC: Hereditary spherocytosis ***35***
TOPIC: Sickle cell disease ***36***
TOPIC: Beta thalassemias ***52***
TOPIC: Aplastic anemia ***53***
TOPIC: Platelets - abnormal count and cancer ***54***
TOPIC: Thrombophilia ***63***
TOPIC: Immune thrombocytopenia (ITP) in adults ***64***
TOPIC: Bruising ***65***
TOPIC: Antiphospholipid syndrome ***74***
TOPIC: DVT/DVT prevention for travelers ***76***
TOPIC: Post-thrombotic syndrome ***80***
TOPIC: Haematological cancers - recognition and referral ***81***
TOPIC: Acute myeloid leukemia ***83***
TOPIC: Chronic lymphocytic leukemia ***85***
TOPIC: Burkitt's lymphoma ***86***
TOPIC: Non-Hodgkin's lymphoma ***89***
TOPIC: Polycythemia/Erythrocytosis ***91***
TOPIC: Von Willebrand's disease ***97***
TOPIC: Haemophilia ***98***
TOPIC: Cryoglobulinaemia ***99***
TOPIC:Tranexamic acid ***100***
TOPIC: MGUS ***101***
TOPIC: Neutropenic sepsis ***102***
TOPIC: Paroxysmal nocturnal haemoglobinuria ***103***

TOPIC: Splenomegaly ***104***
TOPIC: Waldenstrom's macroglobulinaemia ***106***
TOPIC: Wiskott-Aldrich syndrome ***107***

TOPIC: Anemia



TOPIC: Macrocytic anemia

Macrocytosis is defined as a red blood cell (RBC) mean corpuscular volume (MCV) above the upper limit of normal, which is typically >96 to 100 femtoliters (fL) in adults. Macrocytic anemia can be further classified according to the presence of reticulocytes and megaloblastic changes on the peripheral blood smear.

Macrocytic anaemia can be divided into causes associated with a megaloblastic bone marrow and those with a normoblastic bone marrow

Megaloblastic causes	Normoblastic causes	
• vitamin B12 deficiency folate deficiency	e.g. secondary to methotrexate	• alcohol • liver disease • hypothyroidism • pregnancy • reticulocytosis • myelodysplasia • drugs: cytotoxics

Investigations:

- CBC and reticulocyte count
- Serum vitamin B12 and folate level
- Thyroid-stimulating hormone (TSH)
- Liver function tests
- Copper level, especially in individuals who have undergone gastric bypass surgery

TOPIC: Anemia - B12 and folate deficiency

Definition:

- Deficiencies of vitamin B12 or folate are the most frequent causes of megaloblastic anaemia.
    - Megaloblastic anaemia is characterized by larger than normal red blood cells developing in the bone marrow (macrocytosis), with immature nuclei due to defective DNA (deoxyribonucleic acid) synthesis. Macrocytosis is an increase in the mean red cell volume to above the normal range (greater than 100 femtolitres).
- Anaemia is defined by haemoglobin (Hb) levels:
    - In children aged 12–14 years — Hb concentration less than 120 g/L.
    - In men (aged over 15 years) — Hb concentration less than 130 g/L.
    - In women (aged over 15 years) — Hb concentration less than 120 g/L.
    - In women who are pregnant — Hb concentration less than 110 g/L.
        - An Hb level of 110 g/L or more appears adequate in the first trimester, a level of 105 g/L appears adequate in the second and third trimesters.
    - Postpartum — Hb concentration less than 100 g/L.
- There is no gold standard test for measuring vitamin B12 deficiency, but the likelihood of deficiency can be determined by measuring serum cobalamin.
    - A serum cobalamin of less than 200 nanograms/L is sensitive enough to diagnose 97% of people with vitamin B12 deficiency.
- There is no clear consensus on the level of serum folate that indicates deficiency. Conventionally, a serum folate lower than 3 micrograms/L is used as a guideline as the risk of megaloblastic anaemia greatly increases below this level.

Causes:

Vitamin B12 deficiency

- Body stores of vitamin B12 are in the region of 2–5 mg, which is sufficient to last for 2–5 years.
- Vitamin B12 combines with intrinsic factor (IF), which is produced by parietal cells in the stomach, to form an IF–B12 complex. The complex binds to surface receptors for IF in the distal ileum, which then allows absorption of vitamin B12 to take place.
- Pernicious anemia (an autoimmune disorder which results in reduced production of IF) is the most common cause of severe vitamin B12 deficiency.
- Other causes of vitamin B12 deficiency are rare, but include:
    - Drugs — colchicine, metformin, nitrous oxide, protein pump inhibitors, H2-receptor antagonists.
    - Gastric — total or partial gastrectomy, congenital intrinsic factor deficiency or abnormality, Zollinger-Ellison syndrome.
    - Inherited — intrinsic factor receptor deficiency (Imerslund Gräsback syndrome), congenital intrinsic factor deficiency (juvenile pernicious anaemia), cobalamin mutation, transcobalamin deficiency.
    - Intestinal — malabsorption (for example gluten-induced enteropathy), ileal resection, Crohn's disease, blind loop syndrome, parasites (for example, giardiasis, fish tapeworm).
    - Nutritional — malnutrition, vegan diet.

Pernicious anaemia

- Pernicious anaemia is caused by an autoimmune process which affects the gastric mucosa, leading to atrophy. This atrophy reduces the number of parietal cells which secrete intrinsic factor (IF). IF is essential for the absorption of vitamin B12.
    - Antibodies to IF are very specific for pernicious anaemia, however, they are present in only 50% to 70% of people with the condition.
    - Anti-parietal cell antibodies occur in 90% of people with pernicious anaemia, however, they are less specific and common in older people (for example, 16% of women aged over 60 years) who do not have pernicious anaemia.
- Pernicious anaemia affects more women than men, with a ratio of 1.6:1, and a peak occurrence in those aged 60 years.
- People with pernicious anaemia are at increased risk of developing gastric cancer (2–3% of all cases of pernicious anaemia), and there is an association with other autoimmune diseases (including myxoedema, thyrotoxicosis, Hashimoto's disease, Addison's disease, and vitiligo).

Folate deficiency

- Folate is usually absorbed through the upper part of the small intestine and body stores are around 10–12 mg, which is sufficient for around 4 months.
- Folate deficiency is often caused by problems with dietary intake alone, or in a combination with increased folate usage, or malabsorption. Causes can be categorised as due to:
    - Drugs — alcohol, anticonvulsants, nitrofurantoin, sulfasalazine, methotrexate, trimethoprim.
    - Increased requirements, for example, due to:
        - Pregnancy and lactation, prematurity.
        - Malignancy (for example leukaemia, carcinoma, or lymphoma).
        - Blood disorders (for example haemolytic anaemias, sickle cell anaemia, myelofibrosis).
        - Inflammatory diseases (for example tuberculosis, Crohn's disease, or malaria).
        - Exfoliative skin diseases.
    - Excessive urinary excretion (for example due to congestive heart failure, acute liver damage, or chronic dialysis).
    - Liver disease.
    - Malabsorption — due to coeliac disease, tropical sprue, congenital specific malabsorption, jejunal resection, or inflammatory bowel disease.
    - Nutrition — alcoholism or poor diet.

Complications:

- **Complications of anaemia**
    - Adults (especially the elderly) with severe anaemia are at risk of cardiopulmonary complications, such as heart failure.
- **Complications of vitamin B12 deficiency**

    - Neurological symptoms — for example, paraesthesia, ataxia, progressive symmetrical neuropathy which affects the legs more than the arms, numbness, poor motor coordination, memory lapses, and age-related cognitive impairment.
        - Optic atrophy and severe psychiatric symptoms occur rarely.
    - **Note:** neurological disorders can occur independently of the haematological manifestations of pernicious anaemia.
- **Complications of folate deficiency**
    - Maternal folate deficiency in pregnancy is associated with prematurity.
    - Folate deficiency may be associated with cardiovascular disease, and studies have produced conflicting results regarding cancer risk.
- **Complications that can occur with either vitamin B12 or folate deficiency**
    - Vitamin B12 or folate deficiency in pregnancy increases the risk of neural tube defects (such as spina bifida, anencephaly, and encephalocele) in the foetus. Incidence of neural tube defects increases with decreasing serum folate or vitamin B12 levels.
    - Nutritional deficiency of vitamin B12 or folate may cause ineffective production of any types of blood cells derived from bone marrow.
    - Vitamin B12 or folate deficiency may cause sterility. This is reversible with appropriate vitamin supplementation.

Diagnosis:

Assessment:

- Diagnosis of anaemia caused by vitamin B12 or folate deficiency is made through history, examination and investigations.
- Take a detailed medical history, and ask about:
    - Symptoms of anaemia, vitamin B12 and folate deficiency.
    - Possible causes of deficiency:
        - Personal or family history of autoimmune diseases.
        - Diet — vegan or vegetarian.
        - Drug history.
- Examine the person to look for signs of anaemia, B12 and folate deficiency.
- Arrange necessary investigations.
- Consider non-megaloblastic causes of macrocytosis.

Signs and symptoms:

- The main clinical features of megaloblastic anaemia are similar regardless of the underlying cause. This includes whether it is caused by folate deficiency, or vitamin B12 deficiency, except that severe neuropathy does not occur with folate deficiency.
    - The onset of megaloblastic anaemia is usually insidious, with gradually progressive signs and symptoms.
    - Anaemia due to pernicious anaemia may develop gradually over several years, and symptoms may not appear until it is severe.
    - Note: clinical features of vitamin B12 deficiency can occur without anaemia and without low serum levels of vitamin B12.
- **Symptoms of vitamin B12 and folate deficiency include:**

- Cognitive changes.
- Dyspnoea.
- Headache.
- Indigestion.
- Loss of appetite.
- Palpitations.
- Tachypnoea.
- Visual disturbance.
- Weakness, lethargy.

- People with pernicious anaemia may present with symptoms of associated disorders, for example, myxoedema, other thyroid disorders, vitiligo, stomach cancer, or Addison's disease.
- **Signs of vitamin B12 and folate deficiency include:**
  - Anorexia.
  - Angina (in older people).
  - Angular cheilosis.
  - Brown pigmentation affecting nail beds and skin creases (but not mucous membranes).
  - Congestive heart failure (in older people).
  - Episodic diarrhoea.
  - Glossitis — red smooth and shiny tongue, perhaps with ulcers.
  - Heart murmurs.
  - Liver enlargement.
  - Mild jaundice — a lemon-yellow tint.
  - Mild pyrexia.
  - Oropharyngeal ulceration.
  - Pallor of mucous membranes or nail beds.
  - Tachycardia.
  - Weight loss.
- **Neurological complications associated with vitamin B12 deficiency include:**
  - Loss of cutaneous sensation.
  - Loss of mental and physical drive.
  - Muscle weakness.
  - Optic neuropathy.
  - Psychiatric disturbances – these range from mild neurosis to severe dementia.
  - Symmetrical neuropathy affecting the legs more than the arms — this usually presents with ataxia or paraesthesia.
  - Urinary or faecal incontinence.

Investigations:

- **For people with suspected vitamin B12 or folate deficiency, initially arrange:**
  - A full blood count to determine mean cell volume (MCV), haematocrit and haemoglobin levels, and a blood film — which help to identify megaloblastic anaemia.
  - Measurement of serum cobalamin and folate levels — to determine the cause of anaemia.
  - Additional investigations, such as liver function tests, gamma-glutamyl transpeptidase, and/or thyroid function test — may be required depending on clinical judgement to rule out other causes of macrocytic anaemia.

Interpreting investigations results:

- **Full blood count**
  - A high mean cell volume ([MCV] greater than 100 femtolitres) is indicative of macrocytosis, however, MCV may be normal if there is associated iron deficiency, or if anaemia develops more rapidly over the course of a few weeks.
  - A normal MCV does not exclude the need for cobalamin testing, as neurological impairment occurs with a normal MCV in 25% of cases.
  - The white cell count and platelet count may be reduced if the anaemia is severe.
  - There may be a low reticulocyte count in relation to the degree of anaemia (usually 1–3%).
- **Blood film**
  - Hyper-segmented neutrophils (more than 5% of neutrophils with five or more lobes, or one or more neutrophils with 6 or more lobes) and the presence of oval macrocytes, may suggest either vitamin B12 or folate deficiency, but their presence is not sensitive or specific in early cobalamin deficiency.
    - Oval macrocytes, hypersegmented neutrophils and circulating megaloblasts in the blood film, as well as megaloblastic change in the bone marrow, are typical features of clinical cobalamin deficiency.
- **Vitamin B12 level** — interpret the results of the serum cobalamin test taking into account clinical symptoms, other laboratory findings and the following limitations:
  - Cobalamin levels are not easily correlated with clinical symptoms, although people with cobalamin levels of less than 100 nanograms/L (75 picomol/L) usually have clinical or metabolic evidence of vitamin B12 deficiency.
    - In the elderly, low serum cobalamin concentrations usually in the range 100–160 nanograms/L may occur in the absence of anaemia or macrocytosis, and clinically significant vitamin B12 deficiency may be present even with cobalamin levels in the normal range.
  - Women taking oral contraceptives may show decreased cobalamin levels because of a decrease in cobalamin carrier protein, however, this may not result in deficiency.
  - Serum cobalamin levels fall in pregnancy and are less reliable in determining deficiency.
- **Folate level**
  - Serum folate of less than 7 nanomol/L (3 micrograms/L) is used as a guide to indicate folate deficiency.
  - However, there is an indeterminate zone with folate levels of 7–10 nanomol/L (3–4.5 micrograms/L), so low folate should be interpreted as suggestive of deficiency and not diagnostic.

- If there is a strong clinical suspicion of folate deficiency but normal serum levels, red cell folate can be measured once cobalamin deficiency has been ruled out.
    - A red cell folate level below 340 nanomol/L (150 micrograms/L) is consistent with clinical folate deficiency in the absence of vitamin B12 deficiency.

How should I manage people with confirmed vitamin B12 or folate deficiency?

- Determine whether there is an underlying cause for the vitamin B12 or folate deficiency — this may require specialist referral.
    - If folate levels are low, assess dietary folic acid intake, and if history suggests malabsorption, check for coeliac disease by testing for anti-endomysial or anti-transglutaminase antibodies (depending on the local laboratory).
        - The main cause of folate malabsorption is gluten-induced enteropathy.
    - If cobalamin levels are low, check for serum anti-intrinsic factor antibodies.
        - Also check for serum anti-intrinsic factor antibodies in people with strong clinical features of B12 deficiency, such as megaloblastic anaemia or subacute combined degeneration of the cord, despite a normal cobalamin level.
- Determine whether the person has experienced complications of anaemia, vitamin B12 or folate deficiency.
- Refer or treat the person where appropriate depending on the suspected cause.

Differential diagnosis:

- Vitamin B12 and folate deficiencies are not the only causes of macrocytosis.
- Other (non-megaloblastic) causes of macrocytosis include:
    - Alcohol — may cause macrocytosis with neither anaemia nor a change in liver function.
        - Alcohol is the most frequent cause of a raised mean cell volume (MCV) in the absence of anaemia.
    - Drugs — antimetabolites, such as hydroxycarbamide, methotrexate and azathioprine.
    - Haematological abnormalities such as:
        - Myelodysplasia.
        - Aplastic anaemia.
        - Pure red cell aplasia.
        - Plasma protein changes (for example myeloma).
        - Reticulocytosis.
    - Liver disease — chronic liver disease is associated with anaemia that is mildly macrocytic.
    - Pregnancy and the neonatal period.
    - Severe thyroid deficiency — anaemia caused by hypothyroidism is often macrocytic. However, MCV falls once thyroxine treatment is started.
    - Smoking.

Management:

From age 16 years onwards.
How should I treat a person with vitamin B12 deficiency anaemia?

- **For people with neurological involvement**

    - Seek urgent specialist advice from a neurologist and/or haematologist.
    - Ideally, management should be guided by a specialist, but if specialist advice is not immediately available, consider the following:
        - Initially administer hydroxocobalamin 1 mg intramuscularly on alternate days until there is no further improvement, then administer hydroxocobalamin 1 mg intramuscularly every 2 months.
- **For people with no neurological involvement**
    - Initially administer hydroxocobalamin 1 mg intramuscularly three times a week for 2 weeks.
    - The maintenance dose depends on whether the deficiency is diet related or not. For people with B12 deficiency that is:
        - *Not* thought to be diet related — administer hydroxocobalamin 1 mg intramuscularly every 2–3 months for life. Alternatively, daily large oral doses (500 to 1000 micrograms) can be considered for maintenance therapy.
        - Thought to be diet related — advise people either to take oral cyanocobalamin tablets 50–150 micrograms daily between meals, or have a twice-yearly hydroxocobalamin 1 mg injection.
        - In vegans, treatment may need to be life-long, whereas in other people with dietary deficiency replacement treatment can be stopped once the vitamin B12 levels have been corrected and the diet has improved.
    - Give dietary advice about foods that are a good source of vitamin B12 — good sources of vitamin B12 include:
        - Eggs.
        - Foods which have been fortified with vitamin B12 (for example some soy products, and some breakfast cereals and breads) are good alternative sources to meat, eggs, and dairy products.
        - Meat.
        - Milk and other dairy products.
        - Salmon and cod.

How should I treat a person with folate deficiency anaemia?

- **Prescribe oral folic acid 5 mg daily** — in most people, treatment will be required for 4 months.
    - However, folic acid may need to be taken for longer (sometimes for life) if the underlying cause of deficiency is persistent.
- **Check vitamin B12 levels in all people before starting folic acid** — treatment can improve wellbeing, mask underlying B12 deficiency, and allow neurological disease to develop.
- **Give dietary advice about foods that are a good source of folic acid** — good sources of folate include:
    - Asparagus.
    - Broccoli.
    - Brown rice.
    - Brussels sprouts.
    - Chickpeas.
    - Peas.

When should I refer a person with vitamin B12 or folate deficiency anaemia?

- Seek urgent advice from a haematologist if the person has neurological symptoms, or is a pregnant woman.
- **Refer to a haematologist if:**
  - The cause of vitamin B12 or folate deficiency is uncertain, or unclear following investigations.
  - Haematological malignancy or other blood disorder is suspected — refer urgently using a suspected cancer pathway referral (for an appointment within 2 weeks).
  - The person does not respond to treatment.
  - The mean cell volume is persistently greater than 105 femtolitres.
- **Refer to a gastroenterologist if:**
  - Malabsorption or an inflammatory bowel disorder is suspected.
  - The person has pernicious anaemia and gastrointestinal symptoms, especially if there is a suspicion of gastric cancer (for example co-existing iron deficiency).
    - If gastric cancer is suspected, refer urgently using a suspected cancer pathway referral (for an appointment within 2 weeks).
- Consider referral to a dietician if vitamin B12 or folate deficiency is thought to be due to a poor diet.

What monitoring is required after treatment for vitamin B12 or folate deficiency has started?

- **Perform a full blood count and reticulocyte count:**
  - **Within 7–10 days of starting treatment.**
    - A rise in the haemoglobin level and an increase in the reticulocyte count to above the normal range indicates that treatment is having a positive effect.
    - If there is no improvement following initial B12 treatment, check serum folate level (if this has not been done already).
  - **After 8 weeks of treatment.**
    - Blood counts and mean cell volume should have normalised.
    - At this point, also measure iron and folate levels in people being treated for B12 deficiency to ensure that other deficiencies have not been masked.
  - **On completion of folic acid treatment** to confirm a response.
- **Measuring cobalamin levels is unhelpful** as levels increase with treatment regardless of how effective it is, and retesting is not usually required.
  - However, cobalamin can be measured 1–2 months after starting treatment if there is no response.
- **Neurological recovery may take some time** — improvement begins within one week and complete resolution usually occurs between six weeks and three months.
- **Ongoing monitoring is unnecessary** unless a lack of compliance with treatment is suspected, anaemia recurs, or neurological symptoms do not improve, or progress.

Prescribing information:

Cyanocobalamin:

Cautions and contraindications

- There are no listed contraindications or cautions for cyanocobalamin.

Adverse effects

- Adverse effects of cyanocobalamin include:
  - Sensitisation (rarely) — itching exanthema, or exceptionally as anaphylactic shock.
  - Acneform and bullous eruptions.

Drug interactions

- **Absorption of cyanocobalamin from the GI tract may be reduced by:**
  - Aminoglycosides.
  - Aminosalicylic acid.
  - Anticonvulsants.
  - Biguanides.
  - Chloramphenicol.
  - Cholestyramine.
  - Cimetidine.
  - Methyldopa.
  - Omeprazole
  - Potassium salts.

The manufacturer states that the clinical relevance of many of these interactions is likely to be small.

Pregnancy and breastfeeding

- **Pregnancy** — cyanocobalamin can be used in pregnancy to correct an established vitamin B12 deficiency.
- **Breastfeeding** — cyanocobalamin is excreted in breastmilk, but is unlikely to be harmful.

Folic acid:

Cautions and contraindications

- **Do not prescribe folic acid to people with:**
  - Pernicious anaemia or undiagnosed megaloblastic anaemia (unless also prescribing vitamin B12) — use of folic acid alone may precipitate subacute combined degeneration of the spinal cord.
  - Malignant disease (unless megaloblastic anaemia caused by folate deficiency is an important complication).
- Prescribe folic acid with caution to people:
  - Who may have folate dependent tumours.

Adverse effects

- **Rare adverse effects of folic acid include:**
  - Gastrointestinal — abdominal distension, flatulence, anorexia, nausea.
  - Immune system — allergic reactions, including erythema, rash, pruritus, urticaria, dyspnoea, and anaphylactic reactions (including shock).

Drug interactions

- **Drug interactions with folic acid include:**
  - Carbamazepine — levels may be reduced if taken concurrently with folic acid. Monitor carbamazepine concentrations and adjust the dose accordingly.
  - Fluorouracil — levels may be increased producing toxicity. Avoid concurrent use with folic acid.
  - Fosphenytoin — levels may be reduced if taken concurrently with folic acid. Monitor fosphenytoin concentrations and adjust the dose accordingly.
  - Phenobarbital — levels may be reduced if taken concurrently with folic acid. Monitor phenobarbital concentrations and adjust the dose accordingly.
  - Phenytoin — levels may be reduced by 16–50% if taken concurrently with folic acid. Monitor phenytoin concentrations and adjust the dose accordingly.
  - Primidone — levels may be reduced if taken concurrently with folic acid. Monitor primidone concentrations and adjust the dose accordingly.
  - Sulfasalazine — absorption of folic acid may be reduced. Monitor blood counts closely if folic acid is given concurrently with sulfasalazine.
- Be aware that folic acid is removed by haemodialysis.

Pregnancy and breastfeeding

- **Pregnancy** — folic acid can be used during pregnancy.
- **Breastfeeding** — folic acid is actively excreted in breastmilk. No adverse effects of folic acid supplementation have been observed in breastfed infants.

TOPIC: Microcytic anaemia

Microcytosis can be documented from the complete blood count (CBC) and mean cell volume (MCV) MCV <80 femtoliters [fL] for adults or below the age-appropriate value for children or from the peripheral blood smear.

Causes

- Iron-deficiency anaemia
- Thalassaemia
- Congenital sideroblastic anaemia
- Lead poisoning

If history of a normal haemoglobin level associated with a microcytosis. In patients not at risk of thalassaemia, this should raise the possibility of polycythaemia rubra vera which may cause an iron-deficiency secondary to bleeding.

New onset microcytic anaemia in elderly patients should be urgently investigated to exclude underlying malignancy.

TOPIC: Anaemia – iron deficiency:

Definition:

- Iron deficiency occurs as a result of long-term negative iron balance. The iron deficiency spectrum ranges from iron depletion to iron deficiency anaemia.
- Iron deficiency anaemia is defined as diminished red blood cell production due to low iron stores in the body — it is the most common cause of microcytic, hypochromic anaemia, in which the two red cell indices (mean cell volume [MCV] and mean cell haemoglobin [MCH]) are reduced and the blood film shows small (microcytic) and pale (hypochromic) red cells.
- The British Society of Gastroenterology defines anaemia as a Hb concentration below the lower limit of normal for the relevant population, with a laboratory performing the test.
- A serum ferritin level of less than 30 micrograms/L confirms iron deficiency.

Causes:

The cause of Iron deficiency anaemia is often multifactorial, and can be broadly attributed to:

- **Dietary deficiency** — this is rarely a cause on its own in adults. It takes about 8 years for a healthy adult male to develop iron deficiency anaemia due to a poor diet or malabsorption resulting in no iron intake.
- **Malabsorption** — for example, due to coeliac disease, gastrectomy, or Helicobacter pylori infection.
    - Other uncommon gastrointestinal causes include oesophagitis, schistosomiasis or hookworm, and inflammatory bowel disease.
- **Increased loss** — chronic blood loss, especially from the uterus or gastrointestinal tract (GI).
    - In adult men and postmenopausal women, GI blood loss is the most common cause of iron deficiency anaemia. It can be caused by aspirin or nonsteroidal anti-inflammatory drug use, colonic carcinoma, gastric carcinoma, benign gastric ulceration, and angiodysplasia.
    - Menstruation is the most common cause of iron deficiency anaemia in premenopausal women.
    - Other gynaecological causes include haemorrhage in childbirth.
- **Increased requirement** — physiological iron requirements are three times higher in pregnancy than they are in menstruating women, with increasing demand as pregnancy advances.
- **Other causes** — these include blood donation, self-harm, haematuria (rare), nosebleeds (rare), and medication.

Complications:

- **The complications of iron deficiency anaemia can include:**
    - Cognitive and behavioural impairment in children — especially attention deficits.
    - Impaired muscular performance — endurance and exercise capacity are reduced.
    - Heart failure — high-output heart failure can occur in people with severe anaemia, especially in those with a haemoglobin level of less than 50 g/L.
    - Adverse effects on immune status and morbidity from infection (for all age groups).
    - Visual impairment due to retinal haemorrhage (in very severe, rapid onset cases).

- **Iron deficiency anaemia in pregnancy** has been associated with a number of problems, including:
    - Increased morbidity for the mother and infant, and the possibility of lower birthweight.
    - Preterm delivery — increased risk of preterm delivery and perinatal mortality.
    - Maternal postpartum fatigue, altered cognition, and depressive symptoms — this in turn may affect the woman's interactions with the infant.
    - Infant iron deficiency in the first three months of life (by a variety of mechanisms).

Diagnosis:

Assessment:

- Diagnosis of anaemia caused by iron deficiency is made by taking a history, performing an examination, and through investigations.
- **Take a detailed medical history, and ask about:**
    - Symptoms of anaemia:
        - If the anaemia is severe, ask about specific cardiac symptoms (for example angina, palpitations, and ankle swelling).
    - Diet (to identify poor iron intake).
    - Drug history (for example the use of aspirin, nonsteroidal anti-inflammatory drugs, selective serotonin reuptake inhibitors, clopidogrel, corticosteroids, and long-term proton pump inhibitors).
    - A family history of:
        - Iron deficiency anaemia (which may indicate inherited disorders of iron absorption).
        - Bleeding disorders and telangiectasia.
        - Colorectal carcinoma.
        - Haematological disorders (for example thalassaemia).
        - Gastrointestinal disorders.
    - Resectional or bypass surgery of the GI tract including bariatric surgery.
    - A history of overt bleeding, heavy bruising, blood loss during surgery, or blood donation.
    - A history of recent illness that might suggest underlying gastrointestinal bleeding.
    - Menstrual history, pregnancy, or breastfeeding (if appropriate).
    - Travel history (increased risk of hookworm in travelers to the tropics).
    - Weight loss.
    - Exercise regime — iron deficiency anaemia is common in endurance athletes.
- Examine the person to look for signs of anaemia.
- Arrange necessary investigations.
- Consider other causes of anaemia.

Signs and symptoms:

- **Symptoms associated with iron deficiency anaemia depend on how quickly the anaemia develops.**

    - People with chronic, slow blood loss may be able to tolerate very low levels of haemoglobin (for example less than 70g/L) with few symptoms.
    - Fatigue and mild dyspnoea after exertion may be the only symptoms in otherwise healthy people with slow-onset anaemia.
- **Very common symptoms of anaemia include:**
    - Dyspnoea.
    - Fatigue.
    - Headache.
- **Common symptoms of anaemia include:**
    - Cognitive dysfunction.
    - Restless leg syndrome.
- **Rare symptoms of anaemia include:**
    - Dysphagia (in association with oesophageal web which occurs in Patterson-Brown-Kelly or Plummer-Vinson syndromes).
    - Haemodynamic instability.
    - Syncope.
- **Other symptoms include:**
    - Dizziness or lightheadedness.
    - Weakness.
    - Dysgeusia.
    - Irritability.
    - Palpitations.
    - Pica (abnormal dietary cravings, for example for ice or dirt).
    - Pruritus.
    - Sore tongue.
    - Tinnitus.
    - Impairment of body temperature regulation (in pregnant women).
- Serious symptoms such as worsening of pre-existing anginal pain, ankle oedema, or dyspnoea at rest are unlikely unless the haemoglobin level is less than 70g/L, and this indicates additional heart or lung pathology.
    - Angina may occur if there is pre-existing coronary artery disease.
- Symptoms of iron deficiency may occur without anaemia. These symptoms include fatigue, lack of concentration, and irritability.
- **Common or very common signs of iron deficiency include:**
    - Pallor – this may be observed even with mild anaemia.
    - Atrophic glossitis.
    - Dry and rough skin, dry and damaged hair.
    - Diffuse and moderate alopecia.
- **Other signs of iron deficiency include:**

    - Angular cheilosis (ulceration of the corners of the mouth).
    - Nail changes, such as longitudinal ridging and koilonychia (spoon-shaped nails).
    - Worsening of pre-existing tachycardia, murmurs, cardiac enlargement, and heart failure may occur if anaemia is severe (haemoglobin less than 70g/L).
- There may be an absence of signs, even if the person has severe anaemia.

Investigations:

- For people with suspected iron deficiency anaemia, arrange a full blood count (FBC).
- If results of the FBC show a low haemoglobin and low mean cell volume (MCV) check the ferritin level — check the ferritin level in all people with an MCV less than 95 femtolitres.
    - Consider checking ferritin levels for pregnant women, but be aware that results may be less reliable in pregnancy.
- If iron deficiency anaemia is suspected despite serum ferritin results within an expected range, consider diagnostic trials of iron treatment. This should only be considered in premenopausal women with a history of menorrhagia, or pregnant women (if there is no suspicion of coeliac disease).
    - **Note:** a diagnostic trial of iron treatment should not be used for men and postmenopausal women. These people are at higher risk of occult gastrointestinal bleeding and malignancy and further investigations should be arranged.
- It is less clear in which groups of people vitamin B12 and folate levels should also be checked, and when this should be done. Consider this particularly if the person is anaemic and:
    - The anaemia is normocytic with a low or normal ferritin level.
    - There is an inadequate response to iron supplements in proven iron deficiency anaemia and no reason for this (for example poor compliance) is apparent.
    - Vitamin B12 or folate deficiency is suspected (for example due to dietary deficiency, malabsorption, or lack of folate supplementation in pregnancy).
    - The person is in an older age group (more at risk of pernicious anaemia).

Interpreting a full blood count

- Other red blood cell changes associated with iron deficiency include:
    - Reduced mean cell Hb (hypochromia).
    - Increased percentage of hypochromic red cells.
    - Anisocytosis (variation in the size of red blood cells).
    - Poikilocytosis (presence of irregularly shaped red blood cells).

For pregnant women:

- A full blood count is routinely conducted in pregnancy.
- Postpartum, an Hb level of below 100 g/L indicates anaemia.
- If a blood film is arranged, it may confirm the presence of microcytic, hypochromic red cells and characteristic 'pencil cells', however:
    - Hypochromia may also occur in haemoglobinopathies (such as thalassaemia).
    - In pregnancy, a physiological reduction in Hb concentration occurs, which does not represent anaemia. There is an increase in red cell mass and plasma volume; the

plasma volume increases more than the red cell mass, leading to haemodilution and a fall in the haematocrit from 40% to 33% on average.

- Normal pregnancy is also associated with a slight increase in MCV (by approximately 4 femtolitres) and for milder cases of iron deficiency, the MCV may not fall below the normal range.

Interpreting ferritin levels

- Serum ferritin level is the biochemical test that most reliably correlates with relative total body iron stores. Low levels indicate low iron stores, except in women who are in the second or third trimester of pregnancy.
- In all people, a serum ferritin level of less than 30 micrograms/L confirms a diagnosis of iron deficiency.
- Ferritin levels can be difficult to interpret if infection or inflammation is present, as levels can be high even in the presence of iron deficiency.
- Ferritin levels are increased independently of iron status in acute and chronic inflammatory conditions, malignant disease, and liver disease.

  - Expert opinion varies as to the level of ferritin that is diagnostic of iron deficiency anaemia in people with chronic inflammation — people with a ferritin concentration of 50 micrograms/L or higher could still be iron deficient.
  - In chronic kidney disease and other inflammatory conditions, a cutoff of 100 micrograms/L has been advised but also needs to be interpreted in addition to other iron studies.
  - If inflammation is suspected to be spuriously affecting the ferritin result, other markers of inflammation (for example C-reactive protein) or measures of iron status (for example total iron binding capacity) need to be considered. Advice should be sought from haematology or clinical biochemistry if in doubt about the selection of further tests and interpretation of results.
- A serum ferritin concentration of greater than 100 micrograms/L may not rule out iron deficiency anaemia, but a result greater than this level should be interpreted in addition to other iron studies.
  - For people with no known inflammatory states and in whom the ferritin level is indeterminate (31 to 99 micrograms/L) further tests may be required to ascertain iron status.

For pregnant women:

- Serum ferritin level is considered to be a reliable indicator of iron deficiency in the first trimester (in the absence of infection or inflammation).
- A serum ferritin concentration of less than 15 micrograms/L indicates iron depletion in all stages of pregnancy, and treatment should be considered when levels fall below 30 micrograms/L, as this indicates early iron depletion, which will continue to fall unless treated.

When should I consider a diagnostic trial of iron treatment?

- If the diagnosis of iron deficiency anaemia is in doubt despite serum ferritin results, a diagnostic trial of oral iron treatment for 2-4 weeks may be considered in premenopausal women with a history of menorrhagia, or pregnant women (if there is no suspicion of coeliac disease).
  - A diagnostic trial of iron treatment should not be used for men and postmenopausal women — gastrointestinal sources of bleeding need to be excluded.

- A trial of oral iron should be considered as the first-line diagnostic test for normocytic or microcytic anaemia in pregnant women with no haemoglobinopathy.
    - In women with known haemoglobinopathy, serum ferritin should be checked before starting a trial of iron.
    - In women with unknown haemoglobinopathy status, a trial of iron should be offered. However, haemoglobinopathy screening should be undertaken without delay in accordance with the NHS sickle cell and thalassaemia screening programme guideline, but with an awareness that iron deficiency can lower the haemoglobin A2 percentage.

How should I manage people with confirmed iron deficiency?

- For people with a confirmed diagnosis of iron deficiency anaemia, ideally document the likely cause — if there is no obvious cause, appropriate further investigation is generally guided by the person's age and sex.
- **For all people with iron deficiency anaemia:**
    - Screen for coeliac disease using coeliac serology (presence of anti-tissue transglutaminase antibodies).
        - If positive, refer for further investigations.
    - Test the urine for blood.
- Consider stool examination to detect parasites, if appropriate from the person's travel history.
    - **Note:** faecal occult blood testing is not recommended in the investigation of iron deficiency anaemia.
- **It is usually unnecessary to further investigate the following groups of people prior to** treatment**:**
    - Otherwise, healthy young people in whom history clearly suggests a cause — for example, regular blood donors.
    - Menstruating young women with no history of gastrointestinal symptoms or family history of colorectal cancer.
    - Pregnant women — investigations (to determine an underlying cause or the presence of complications) are not usually needed if anaemia develops during pregnancy, unless:
        - The anaemia is severe.
        - The history and examination suggest an alternative cause of iron deficiency (for example inflammatory bowel disease).
        - There is no response to iron supplementation.
    - People who are terminally ill or unable to undergo invasive investigations — the appropriateness of investigating people with severe comorbidity (or, in some circumstances, advanced age), especially if management would not be influenced by the results, should be discussed with the person and their family and carers.
    - People who refuse further investigations.
- Refer people where appropriate, or treat people depending on the underlying cause.

Differential diagnosis:

- The differential diagnosis of microcytic anaemia includes:
  - **Thalassaemia** — for people with thalassaemia trait (alpha or beta), the mean cell volume (MCV) and mean cell haemoglobin (MCH) concentration are all reduced and are very low for the degree of anaemia.
  - **Sideroblastic anaemias** (very rare) — alcoholism can be a cause of reversible sideroblastic anaemia. Hepatosplenomegaly is found in a third to a half of people with sideroblastic anaemia and is not present in iron deficiency anaemia.
  - **Anaemia of chronic disease** — on analysis of the full blood count (FBC), in 80% of cases, anaemia of chronic disease is normocytic and normochromic. However, in 20% of cases, it can present as microcytic, hypochromic anaemia like iron deficiency anaemia.
  - **Lead poisoning** (rare in adults) — people may have a history of risk factors, such as occupational exposures (for example, exposure to lead paint).
- Thalassaemia and sideroblastic anaemia are both associated with an accumulation of iron, so tests will show an increase in serum iron and ferritin, and a low total iron-binding capacity.

Management:

From age 12 years onwards.
When should I refer someone with iron deficiency anaemia?

- **Urgently refer people with iron deficiency anaemia using a suspected cancer pathway for an appointment within 2 weeks if they are:**
  - Aged 60 years or over.
- **Consider an urgent referral for people with iron deficiency anaemia using a suspected cancer pathway for an appointment within 2 weeks if they are:**
  - Aged under 50 years and present with rectal bleeding.
- **Refer to gastroenterology:**
  - **All men and postmenopausal women with iron deficiency anaemia unless they have overt non-gastrointestinal bleeding.**
    - Men with a haemoglobin (Hb) level less than 120 g/L and postmenopausal women with an Hb level less than 100 g/L should be investigated more urgently, as lower levels of Hb suggest more serious disease.
  - **All people aged 50 years or over** with marked anaemia, or a significant family history of colorectal carcinoma, even if coeliac disease is found.
  - **Premenopausal women if they are aged under 50 years** and have colonic symptoms, a strong family history (two affected first-degree relatives or just one first-degree relative affected before the age of 50 years) of gastrointestinal cancer, persistent iron deficiency anaemia despite treatment, or if they do not menstruate (for example following hysterectomy).
- **Refer women to gynaecology if:**
  - Menorrhagia is unresponsive to medical management.
  - There is postmenopausal bleeding.
    - For women aged over 55 years — refer urgently using a suspected cancer pathway for an appointment within 2 weeks.
    - For women aged under 55 years — consider referring urgently using a suspected cancer pathway for an appointment within 2 weeks.

- **Refer pregnant women to obstetrics** if there are significant symptoms and/or severe anaemia (haemoglobin less than 70 g/L), if pregnancy is at advanced gestation (over 34 weeks), or if there is failure to respond to a trial of oral iron.
- **Also arrange referral:**
    - If coeliac serology is positive — refer to gastroenterology.
    - If the person has profound anaemia with signs of heart failure.
    - If the person is unable to tolerate, or is not responding to oral iron treatment.
    - If the person has initially responded to iron treatment but develops anaemia again without an obvious underlying cause.
    - When the type of anaemia is in doubt.
    - When further haematological investigations are required that cannot be carried out in primary care (such as bone marrow examination or an investigation of bleeding state).
- **Note:** treatment **with iron replacement therapy should not be deferred while awaiting investigations for iron deficiency anaemia, unless colonoscopy is imminent.**

How should I treat a person with iron deficiency anaemia?

- Address any underlying causes that can be managed in primary care (for example treat menorrhagia or stop nonsteroidal anti-inflammatory drugs, if possible).
    - If dietary deficiency of iron is thought to be a contributory cause of iron deficiency anaemia, advise the person to maintain an adequate balanced intake of iron-rich foods (for example dark green vegetables, iron-fortified bread, meat, apricots, prunes, and raisins) and consider referral to a dietitian.
- Prescribe all people with iron deficiency anaemia one tablet once daily of oral ferrous sulfate, ferrous fumarate, or ferrous gluconate — continue treatment for 3 months after the iron deficiency is corrected to allow stores to be replenished.
    - If this is not tolerated, reduce the dose to one tablet on alternate days, or consider alternative oral preparations.
    - Consider parenteral iron if oral iron is contraindicated, ineffective or not tolerated.
    - Do not wait for investigations to be carried out before prescribing iron supplements.
- Monitor the person to ensure that there is an adequate response to iron treatment.
- Refer the person where appropriate.

Iron supplements

- The aim of treatment is to restore haemoglobin levels and red cell indices to normal, and to replenish iron stores.
- A dose of 65 mg elemental iron (ferrous sulfate 200 mg) once daily (on an empty stomach) is needed to treat iron deficiency anaemia. Dose-related adverse effects (gastrointestinal disturbance) from taking an iron supplement are commonly experienced.
    - For people with significant intolerance to oral iron replacement therapy options include alternate day dosing, oral ferric maltol, or parenteral iron preparations.
    - **Note:** switching to an alternate oral iron salt is not supported by evidence.

What information and advice should I give people taking iron supplements?

- Advise people taking iron supplements that:

    - They may experience adverse effects, but these usually settle down with time — stress the importance of compliance with treatment.
    - If they do experience gastrointestinal adverse effects, discomfort may be minimized by taking the iron supplement with or after food, or reducing the dose to alternate days.
        - If these measures fail, they should return to discuss other treatment options.
- Explain the monitoring requirements.
- Advise people (especially the parents of young children) about the safe storage of iron supplements, as accidental overdose can be fatal.

What monitoring is recommended for someone being treated for iron deficiency anaemia?

- Recheck haemoglobin levels (full blood count) within the first 4 weeks of iron supplement treatment to assess the person's response. The haemoglobin concentration should rise by about 20 g/L over 3–4 weeks.
    - If there is a response, check the full blood count at 2–4 months to ensure that the haemoglobin level has returned to normal.
- Once haemoglobin concentration and red cell indices are normal:
    - Continue iron treatment for 3 months to aid replenishment of iron stores, and then stop.
    - Then monitor the person's full blood count periodically — for example, at 3, 6, 12, and 24 months.
- If haemoglobin or red cell indices drop below normal, prescribe iron supplements.
    - Further investigation is only necessary if haemoglobin or red cell indices cannot be maintained this way or if there is any evidence of an active undiagnosed pathology (for example ongoing weight loss or chronic unexplained diarrhoea, persistently elevated inflammatory markers and the persistence or recurrence of IDA).
    - Consider an ongoing prophylactic dose in people who are at particular risk of iron deficiency anaemia.

Ongoing iron supplementation

- An ongoing prophylactic dose of iron (200 mg ferrous sulfate daily) may be beneficial in some people who have:
    - Recurring anaemia (such as in an elderly person) and further investigations are not indicated or appropriate.
    - A diet which is unlikely to meet daily iron intake recommendations — for example, some plant-based diets.
    - Malabsorption — for example, coeliac disease.
    - Menorrhagia.
    - Had a gastrectomy.
- Ongoing prophylaxis may also be beneficial for:
    - Pregnant women.
    - People undergoing haemodialysis.

How should I manage someone if there is inadequate response to initial iron treatment?

- **Assess compliance** and whether the iron treatment is tolerated — if an oral iron supplement (usually ferrous sulfate) is not tolerated, address the adverse effects:
    - Offer a laxative to people with constipation.
    - Offer reassurance to people who have black stools.
    - Recommend the person takes iron with or after meals.
    - Reduce the dose frequency of the iron supplement to alternate days.
    - Consider alternative oral preparations.
- If the person is still unable to tolerate oral iron supplements, seek specialist advice.
- **Refer people for specialist assessment** if there is a lack of response (that is, an increase of less than 20 g/L in the haemoglobin level) after 2–4 weeks.
    - If the person has already had normal upper and lower gastrointestinal investigations for iron deficiency anaemia and the anaemia persists or recurs, consider testing for Helicobacter pylori, and eradicate if present.

Prescribing information:

Iron content in different iron salts

Table 1. Iron content of oral iron preparations

Iron salt	Dose	Elemental iron
Ferrous fumarate	200 mg	65 mg
Ferrous gluconate	300 mg	35 mg
Ferrous sulfate	300 mg	60 mg
Ferrous sulfate, dried	200 mg	65 mg
Ferric maltol	30 mg	30 mg
Sodium feredate	190 mg/5 mL elixir	27.5 mg/5 mL elixir

Adverse effects:

- Adverse effects of iron supplements include:
    - Constipation.
    - Diarrhoea.
    - Epigastric pain.
    - Faecal impaction.
    - Gastrointestinal irritation.
    - Nausea.
- Adverse effects are dose-related and are directly related to the amount of iron absorbed (although the relationship between constipation or diarrhoea is less clear than for nausea and epigastric pain).

    - The incidence of adverse effects is no greater with ferrous sulfate than with other iron salts.
- Adverse effects of oral iron supplements are a common cause of non-compliance with treatment — 10–20% of people are thought to discontinue iron supplements because of adverse effects.

Drug interactions:

- **Oral iron supplements reduce the absorption of some drugs if taken concurrently, including:**
    - Tetracyclines.
    - Quinolones.
    - Bisphosphonates.
    - Zinc.
- **The absorption of oral iron is reduced if taken concurrently with:**
    - Zinc or magnesium salts (for example in antacids).
    - Calcium (for example in milk and dairy products).
    - Tannins (for example in tea, coffee, and cocoa).
    - Phytates (present in cereal grains, legumes, nuts, and seeds).
- **Interactions may occur with the following drugs if taken concurrently with oral iron supplements:**
    - Methyldopa — the antihypertensive effect can be reduced.
    - Levodopa — the bioavailability may be reduced.
    - Levothyroxine — the effects of levothyroxine may be reduced.
    - Penicillamine — the absorption may be reduced by as much as two-thirds.

TOPIC: Normocytic anaemia

Normocytic anemia (MCV between 80 and 100 fL).

Causes of normocytic anemia include:

- anemia of chronic disease
- chronic kidney disease
- aplastic anemia
- hemolytic anemia
- acute blood loss

Investigations:

- CBC and reticulocytes
  - Pancytopenia raises concern for an underlying bone marrow disorder
  - High reticulocyte count suggests hemolysis or blood loss
- RFT
- LFT
- Peripheral blood smear

Management:

Treat the underlying cause

TOPIC: Lead poisoning

Along with acute intermittent porphyria, lead poisoning should be considered in questions giving a combination of abdominal pain and neurological signs. Lead poisoning results in defective ferrochelatase and ALA dehydratase function.

Features

- abdominal pain
- peripheral neuropathy (mainly motor)
- neuropsychiatric features
- fatigue
- constipation
- blue lines on gum margin (only 20% of adult patients, very rare in children)

Investigations

- the blood lead level is usually used for diagnosis. Levels greater than 10 mcg/dl are considered significant
- full blood count: microcytic anaemia. Blood film shows red cell abnormalities including basophilic stippling and clover-leaf morphology
- raised serum and urine levels of delta aminolaevulinic acid may be seen making it sometimes difficult to differentiate from acute intermittent porphyria
- urinary coproporphyrin is also increased (urinary porphobilinogen and uroporphyrin levels are normal to slightly increased)
- in children, lead can accumulate in the metaphysis of the bones although x-rays are not part of the standard work-up

Management - various chelating agents are currently used:

- dimercaptosuccinic acid (DMSA)
- D-penicillamine
- EDTA
- Dimercaprol

- **Sources of lead exposure**

Occupational	**Homes/Buildings**
Plumbers, pipe fitters	Lead-containing paint/pigment
Lead miners	Soil/dust near lead industries, roadways, lead-painted homes
Lead smelters and refiners	Plumbing leachate
Painters	Ceramic ware (especially imported)
Auto repairers	Leaded gasoline
Glass manufacturers	Vinyl miniblinds*
Shipbuilders	**Hobbies and related activities**
Printers	Glazed pottery making
Plastic manufacturers	Target shooting at firing ranges
Police officers	Lead soldering (eg, electronics)
Steel welders or cutters	Painting
Construction workers (especially renovation and rehabilitation)	Preparing lead shot, fishing sinkers
Rubber product manufacturers	Stained-glass making
Gas station attendants (past exposure)	Car or boat repair
Battery manufacturers	Home remodeling
Battery recyclers	**Other sources**
Bridge reconstruction workers	Folk remedies (Mexican: azarcon, greta; Asian: ba-baw-san, bali goli)
Firing range instructors	Tobacco smoking
	Cosmetics
	Moonshine whiskey
	Gasoline "huffing"
	Ayurvedic medications

TOPIC: Autoimmune haemolytic anaemia

Autoimmune haemolytic anaemia (AIHA) may be divided in to 'warm' and 'cold' types, according to at what temperature the antibodies best cause haemolysis. It is most commonly idiopathic but may be secondary to a lymphoproliferative disorder, infection, drugs, or SLE.

Symptoms:
- Hx of anemia
- Jaundice
- +/- hemoglobinuria

Investigations :
- CBC: anaemia, reticulocytosis
- low haptoglobin
- raised lactate dehydrogenase (LDH) and indirect bilirubin
- blood film: spherocytes and reticulocytes
- positive direct antiglobulin test (Coombs' test).

Warm AIHA
Warm is the most common type of AIHA in which the antibody (usually IgG) causes haemolysis best at body temperature (37°C) and haemolysis tends to occur in extravascular sites e.g. the spleen.

Causes:
- idiopathic
- autoimmune disease
- neoplasia
  - lymphoma
  - chronic lymphocytic leukaemia
- drugs: e.g. methyldopa

Management
- treatment of any underlying disorder
- steroids (+/- rituximab) are generally used first-line

COLD AIHA
The antibody in cold AIHA is usually IgM and causes haemolysis best at 4°C. Haemolysis is mediated by complement and is more commonly intravascular. Features may include symptoms of Raynaud's, acrocynaosis and Livido reticularis. Patients respond less well to steroids

Causes of cold AIHA
- neoplasia: e.g. lymphoma
- infections: e.g. mycoplasma, EBV

Management:
- Treating the underlying cause
- Keeping the patient warm and avoiding exposure to cold fluids.
- Depending on the severity patients may require transfusion, pharmacologic therapy, and/or plasma exchange.

TOPIC: Fanconi anaemia

Fanconi anemia (FA) is a rare inherited bone marrow failure syndrome (Autosomal recessive) characterized by pancytopenia, predisposition to malignancy, and characteristic physical abnormalities/congenital malformations. FA is caused by mutations in one of numerous genes involved with DNA repair.

Symptoms:
The condition tends to cause:

- Congenital dysmorphic features
  - Short stature
  - Thumb/radius abnormalities
- Café au lait spots
- CNS involvement
- Pancytopenic bone marrow failure.
- Susceptibility to cancer:
  - Acute myeloid leukaemia (AML).
  - Solid tumours - head and neck squamous cell carcinoma and gynaecological cancer.

Investigations:

- FBC may show macrocytosis with mild anaemia through to pancytopenia. Initial presentation may be with isolated thrombocytopenia or leukopenia.
- Bone marrow biopsy/aspiration reveals progressively hypocellular marrow with loss of myeloid and erythroid precursors and megakaryocytes, eventually becoming typical of aplastic anaemia with fatty marrow.
- Gold standard: chromosomal studies.

Management:

- Families should be offered genetic counselling and cytogenetic testing.
- Blood transfusions are used to maintain counts and treat symptomatic problems.
- Bone marrow/haematopoietic stem cell transplantation using related donors (where possible) is the only curative treatment but the high risk of solid tumors remains.

TOPIC: G6PD deficiency

The enzyme glucose-6-phosphate dehydrogenase (G6PD) is one of the enzymes of the pentose phosphate pathway. This pathway is involved in keeping an adequate amount of the coenzyme nicotinamide adenine dinucleotide phosphate (NADPH) in cells. NADPH in turn maintains the levels of glutathione which protects the red cell from oxidative damage. G6PD is the rate-limiting enzyme in the pentose phosphate pathway. Thus, deficiency of the G6PD enzyme results in reduced glutathione making the red cells vulnerable to oxidative damage and thus liable to haemolysis.

Glucose-6-phosphate dehydrogenase (G6PD) deficiency is the commonest red blood cell enzyme defect. It is more common in people from the Mediterranean and Africa and is inherited in an X-linked recessive fashion. Males are more likely to be affected, and heterozygous females are typically unaffected carriers

Symptoms:

- neonatal jaundice is often seen
- intravascular haemolysis
- gallstones are common
- splenomegaly may be present
- Heinz bodies on blood films. Bite and blister cells may also be seen

Diagnosis:
Using a G6PD enzyme assay. Levels should be checked 3 months after an acute episode of hemolysis, RBCs with the most severely reduced G6PD activity will have hemolysed → reduced G6PD activity → not be measured in the assay → false negative results

Factors that precipitate haemolytic crises

- Certain drugs
  - anti-malarials: primaquine
  - Nitrofurantoin and quinolones including ciprofloxacin, moxifloxacin, nalidixic acid, norfloxacin, and ofloxacin
  - sulph- group drugs: sulphonamides, sulphasalazine, sulfonylureas
- Certain foods (eg, eating fava beans).
- Severe infection.
- Diabetic ketoacidosis
- Acute kidney injury

Some drugs thought to be safe:

- penicillins
- cephalosporins
- macrolides
- tetracyclines
- trimethoprim

Investigation :

- FBC - anaemia.
- Macrocytosis - due to reduced folic acid which is required for erythropoiesis.
- Reticulocyte count - raised; gives indication of the bone marrow activity.
- Blood film - acute haemolysis from G6PD deficiency can produce Heinz bodies, which are denatured haemoglobin and bite cells.
- Haemolysis - reduced levels of haptoglobin and elevated levels of bilirubin; haemoglobinuria.
- G6PD enzyme activity - is the definitive test
- Performing assays for G6PD during haemolysis and reticulocytosis may affect levels and not reflect baseline values.

Management:

- Avoidance of substances that may precipitate haemolysis.
- If haemolysis is marked there may be benefit from folate supplementation.

	G6PD deficiency	Hereditary spherocytosis
Gender	Male (X-linked recessive)	Male + female (autosomal dominant)
Ethnicity	African + Mediterranean descent	Northern European descent
Typical history	• Neonatal jaundice • Infection/drugs precipitate haemolysis • Gallstones	• Neonatal jaundice • Chronic symptoms although haemolytic crises may be precipitated by infection • Gallstones • Splenomegaly is common
Blood film	Heinz bodies	Spherocytes (round, lack of central pallor)
Diagnostic test	Measure enzyme activity of G6PD	EMA binding

TOPIC: Hereditary spherocytosis

Basics:

- most common hereditary hemolytic anemia in people of northern European descent
- Autosomal dominant defect of red blood cell cytoskeleton
- The normal biconcave disc shape is replaced by a sphere-shaped red blood cell
- Red blood cell survival reduced as destroyed by the spleen

Presentation:

- Failure to thrive
- Jaundice, gallstones
- Splenomegaly
- Aplastic crisis precipitated by parvovirus infection
- Degree of haemolysis variable
- MCHC elevated

Initial testing:

- **All patients** — Some aspects of the initial evaluation differ in neonates versus older children and adults since affected neonates tend to have more severe disease and less useful laboratory parameters.

- However, the following are appropriate in virtually all individuals:

- ●**CBC and RBC indices** – All individuals with suspected HS based on family history, neonatal jaundice, or other findings should have a complete blood count (CBC) with reticulocyte count and red blood cell (RBC) indices. The mean corpuscular hemoglobin concentration (MCHC) is often the most useful parameter for assessing spherocytosis; an MCHC ≥36 g/dL is consistent with spherocytes. A low mean corpuscular volume (MCV) is also helpful in some cases, especially in neonates, but variable degrees of reticulo-cytosis make the MCV less useful in older children and adults.
- ●**Blood smear review** – All individuals with suspected HS should have a blood smear reviewed by an experienced individual. RBC parameters to be assessed include the presence and abundance of spherocytes, other abnormal RBC shapes, and the degree of polychromatophilia, which reflects reticulocytosis.
- ●**Hemolysis testing** – Testing for hemolysis is also appropriate in all patients. This includes lactate dehydrogenase (LDH), indirect bilirubin, haptoglobin, and reticulocyte count. Findings consistent with hemolysis include increased LDH and indirect bilirubin, decreased or absent haptoglobin, and an elevated reticulocyte count.
- ●**Coombs testing** – If hemolysis is present, Coombs testing (also called direct antiglobulin testing [DAT]) is usually done to eliminate the possibility of immune-mediated hemolysis, which may be due to hemolytic disease of the fetus and newborn (HDFN) in neonates or autoimmune hemolytic anemia (AIHA) in older children and adults. The results of testing may also be useful to the transfusion service if transfusion is indicated. Coombs testing in HS is negative.

Confirmatory tests EMA binding is our preferred test (eosin-5-Maleimide binding test)

Management:

- acute haemolytic crisis:
  - treatment is generally supportive
  - transfusion if necessary
- longer term treatment:
  - folate replacement
  - splenectomy

Considerations related to splenomegaly – There are no special restrictions (eg, no activity limitations) on children with splenomegaly due to HS. However, those with an unusually large spleen should be made aware of the risk of splenic rupture during contact sports.

TOPIC: Sickle cell disease

Definition:

- **Sickle cell disease encompasses a group of inherited conditions of sickle haemoglobin.**
  - Sickle haemoglobin has an abnormal beta-globin chain which causes it to polymerize when deoxygenated distorting the erythrocyte into a sickle shape.
  - The deformed erythrocytes form clusters which block blood vessels; damage large and small blood vessels; are sequestered in the liver and spleen; and cause anaemia (of varying degrees), intense pain (known as sickle cell crisis), infections, and other complications of sickle cell disease.
- **People with sickle cell disease have inherited the gene for sickle haemoglobin (Hb S) from one parent and a gene for an abnormal haemoglobin variant from the other parent.**
  - If the second abnormal gene is also for Hb S, the person is said to have homozygous sickle cell disease (Hb SS), commonly called sickle cell anaemia. This is the most common and severe type of sickle cell disease.
  - If the second abnormal gene is for a different abnormal haemoglobin, the person is said to have compound heterozygous sickle cell disorder.
  - Heterozygous sickle cell disease can also arise when people inherit a Hb S gene from one parent and a beta-thalassaemia (Hb β thalassaemia) gene from the other parent, causing sickle beta-thalassaemia. A mutation of the normal beta-globin chain blocks production of the normal beta-globin chain (β0), or reduces its production (β+).
- **People who inherit a gene for a sickle haemoglobin (Hb S) from one parent and a gene for normal haemoglobin (Hb A) from the other parent have sickle cell traits (previously known as sickle cell carrier), and their genotype is Hb AS.**
  - These people rarely have symptoms. However, they have a 50% chance of passing the sickle cell gene to their child. If both parents are carriers, there is a 1 in 4 chance that their child will be born with sickle cell disease.

Prognosis:

- Sickle cell disease is variable in its severity and the onset of acute and chronic complications is unpredictable — in its most severe form it causes significant morbidity and mortality.
  - Sickle cell anaemia (Hb SS) and Hb S/β0 thalassaemia are the most severe, but there is variation in clinical phenotype between individuals with these genotypes.

Complications:

- **Acute complications of sickle cell disease are due to the consequences of the 'sickling' of erythrocytes,** which is more likely to happen in certain conditions, such as:
  - Acidosis.
  - Infections.
  - Psychological stress.
  - Extreme exercise.
  - Cold temperatures.
  - Pregnancy.
  - Dehydration.

- Reduced oxygen in the blood (for example due to exertion, anaesthesia, and high altitude).

- **Sickle cell disease may also result in** complications during pregnancy**.**

What are the acute complications of sickle cell disease?

Acute complications of sickle cell disease include:

- **Acute painful crisis**
  - Also known as vasculo-occlusive crisis, it is the most common complication of sickle cell disease in all age groups and can affect any part of the body, but most commonly involves the limbs and back.
  - It may present as skeletal pain due to bone infarction or avascular necrosis, most commonly affecting the femoral head or head of the humerus.
  - Pain is thought to be caused by vaso-occlusion by sickled erythrocytes, leading to ischaemic tissue damage and subsequent inflammation and pain.
  - It may be precipitated by cold, dehydration, prolonged exertion, infection, or ischaemia.
- **Abdominal complications**
  - Abdominal pain is common and may be due to complications — the differential diagnosis includes sequestration syndromes, mesenteric syndrome, constipation, gallstone complications, infective aetiologies (such as pyelonephritis, intra-abdominal abscesses, and diverticulitis), and dysmenorrhoea.
  - Gallstones are common and often cause intermittent abdominal pain.
- **Anaemia**
  - A rapid, significant fall in haemoglobin, usually of at least 20 g/L, may result in the individual becoming symptomatic and major reductions may lead to cardiovascular compromise.
  - Anaemia may be due to haemolysis (for example, due to a transfusion reaction, infection or glucose-6-phosphate dehydrogenase [G6PD] deficiency), reduction in erythropoiesis, blood loss, and sequestration in the spleen and less commonly in the liver.
  - Transient red cell aplasia (most commonly due to parvovirus B19 infection) causes an arrest in erythropoietic maturation — symptoms may include fever, headache, myalgia, arthralgia, and respiratory and gastrointestinal symptoms.
  - Acute splenic sequestration occurs when large numbers of erythrocytes are trapped in the spleen, resulting in splenomegaly and profound anaemia.
  - It can develop rapidly and is more common in children than in adults, although it is occasionally seen in people with milder forms of sickle cell disease.
  - The typical clinical features of acute hepatic sequestration are pain in the right hypochondrium, a tender and enlarged liver, pallor, and circulatory collapse (infrequent).
- **Acute chest syndrome (ACS)**
  - This is a form of acute lung injury that can be severe and life-threatening. It is particularly common in early childhood and can be clinically indistinguishable from pneumonia.

- It is characterized by respiratory symptoms, such as chest pain, tachypnoea, cough, shortness of breath, and/or respiratory distress, generally with fever, in the presence of a new infiltrate on chest X-ray.
- Possible causes include infection, infarction (due to fat emboli from bone infarcts) or a combination of both.

- **Infections**
  - People with sickle cell disease are susceptible to a range of bacterial and other infections including invasive pneumococcal disease, which is the most significant cause of infection-related morbidity and mortality. Increased susceptibility is related to splenic dysfunction.
- **Osteomyelitis**
  - This is one of the most common infectious complications in people with sickle cell disease, and it occurs in people of all ages and genotypes.
  - The commonest sites are the femur, tibia, and humerus and it is most commonly caused by Salmonella species, Gram-negative enteric bacteria, and Staphylococcus aureus.
  - There may be pain, localized warmth, swelling, tenderness, and fever, so it can be difficult to distinguish from vasculo-occlusion.
- **Priapism:**
  - Priapism is very common and has a lifetime incidence of 35–90% in males with sickle cell disease with the majority of first episodes occurring before the age of 20 years.
  - It is caused by obstruction to the venous drainage of the penis due to vaso-occlusion.
  - Males with a sustained and painful erection lasting for more than 1 hour should be encouraged to attend hospital. A priapic episode lasting more than 3 hours is classed as a surgical emergency.
- **Acute kidney injury:**
  - Some people with sickle cell disease develop chronic kidney disease and are at increased risk of acute kidney injury.
  - It occurs in around 2–10% of people admitted with sickle cell anaemia and is more common in those with acute chest syndrome.
  - Acute kidney injury can be caused by dehydration, sepsis, or drugs, or it may occur during general multi-organ failure.
- **Neurological complications:**
  - Central nervous system complications cause significant morbidity and mortality.
  - Acute presentations can include headache, seizures, focal neurological signs, visual impairment, altered consciousness, and acute deterioration in cognition — causes include infection and stroke.
  - Haemorrhagic stroke also occurs at increased rates in adults with sickle cell disease. Hypertension and multi-organ failure are risk factors for cerebral haemorrhage, and aneurysms (reported in over 10% of adults with sickle cell disease) can rupture leading to subarachnoid haemorrhage. Low steady-state haemoglobin concentration, high steady-state white cell-count and transfusion within the previous 14 days are risk factors for haemorrhagic stroke.

What are the chronic complications of sickle cell disease?

Chronic complications of sickle cell disease increase with age and include:

- **Hepatobiliary complications**
  - Hepatobiliary complications in sickle cell disease are common and have a multifactorial aetiology — they may be caused by the sickling process itself, by hypoxic injury due to vaso-occlusion, or by iron overload secondary to multiple blood transfusions.
  - Mild abnormalities in liver function tests are common and may be seen in uncomplicated vaso-occlusive crises, but some patients develop progressive liver disease and end-stage liver disease.
- **Pain**
  - Pain may be due to leg ulcers, tissue damage, avascular necrosis (particularly of the femoral head), or chronic osteomyelitis.
  - Osteopenia may occur due to hyperplasia of the bone marrow.
  - In some people, there is no obvious cause for the pain. Such pain has some features of neuropathic pain and may be related to increased central sensitization to painful stimuli.
- **Anaemia**
- **Respiratory complications**
  - **Pulmonary hypertension**
    - Pulmonary hypertension is common, occurring in 30–40% of adults with sickle cell disease and is associated with increased morbidity and mortality.
    - Symptoms may be nonspecific and include fatigue, limited exercise tolerance, progressive evidence of dyspnoea on exertion (particularly worsening on walking up an incline or climbing stairs), chest pain, light-headedness, and syncope. Sudden death is common.
    - It is associated with increasing age, poor functional capacity, prior history of leg ulceration, renal insufficiency, and markers of haemolysis.
    - Multiple mechanisms have been postulated to contribute to the development of pulmonary hypertension, including chronic intravascular haemolysis leading to reduced nitric oxide scavenging, thrombosis, hypoxaemia, oxidant stress, and asplenia.
  - **Chronic sickle lung**
    - This is an interstitial abnormality of lung parenchyma characterized by progressively worsening restrictive lung disease, pulmonary hypertension, hypoxaemia, and chest pain and with fibrotic changes on CT scan.
    - It affects around 5% of people with sickle cell disease with an average age of onset of 25–33 years, and may develop as a result of recurrent episodes of acute chest syndrome (ACS), but it may also occur in people without a history of ACS.
  - **Sleep-disordered breathing**
    - This is common and includes a group of conditions characterized by complete or partial cessation of normal respiration during sleep.

        - It results in nocturnal hypoxia or obstructive sleep apnoea (OSA) — these are correlated with morbidity including frequent painful vaso-occlusive crisis, risk of future central nervous system (CNS) event, and priapism.

- **Cognitive impairment**
    - Around 20% of children with sickle cell disease have silent cerebral infarcts visible on magnetic resonance imaging (MRI). These are due to very small white-matter lesions and are a risk factor for an overt stroke. The silent infarcts are associated with mild cognitive impairment that may be detected by using neurocognitive tests.
    - Cognitive impairment due to silent infarcts may also occur in adults.
- **Eye complications — sickle retinopathy**
    - This is the most important complication as it can lead to significant visual impairment.
    - It is classified into non-proliferative (NPSR) and proliferative (PSR) forms of retinopathy, although NPSR does not normally lead to visual loss.
    - Mechanical obstruction of retinal capillaries by sickled erythrocytes and direct endothelial damage may also occur.
    - Ischaemic changes result in proliferative retinopathy which is characterized by the development of peripheral retinal neovascularization.
- **Gallstones**
    - Gallstones are common in sickle cell disease, but are more common in sickle cell anaemia.
    - They are usually asymptomatic and are often detected incidentally on routine imaging, but complications may develop due to infection and inflammation of the gall bladder and biliary duct (acute cholecystitis, gallbladder empyema, ascending cholangitis) or due to obstruction of the biliary ducts and acute pancreatitis.
    - Symptomatic biliary tract disease occurs in around 20% of people with sickle cell disease.
- **Impaired nutrition and growth in children**
    - Impaired growth is often apparent after the age of 6 months and may be due to decreased absorption of nutrients, an increased metabolic rate, and/or poor appetite caused by recurrent febrile and painful episodes.
    - Puberty is often delayed by about 2–3 years in children with sickle cell anaemia and 6 months in those with Hb SC.
    - Children on long-term transfusion programmes may have significant iron overload and develop pituitary and/or primary gonadal deficiencies.
- **Leg ulcers**
    - Leg ulcers are a frequent and disabling complication and they may persist for months or years.
    - They are more common in people with higher levels of haemolysis. Mechanical obstruction, high blood viscosity, venous incompetence, hypercoagulability, and thrombosis may also be involved.
    - Leg ulceration is less common in people aged under 20 years and is more frequent in males and in people with sickle cell anaemia.
    - They typically appear just above the ankle and are often bilateral, and can be very painful.

- **Renal complications**
  - Renal complications occur in around 60% of people with the more severe forms of sickle cell disease at some point in their lives.
  - Sickle nephropathy presents as a spectrum from painless haematuria, proteinuria, and progressive loss of function to end-stage renal disease.
  - A reduction in medullary blood flow due to ischaemia, an increase in glomerular blood flow, and papillary necrosis all contribute to kidney damage.
  - Sickle cell obstruction of small vessels leads to medullary insufficiency, which causes large quantities of dilute urine to be produced. This makes the person susceptible to dehydration. Nocturnal enuresis is common in all children, but there is an increased rate in children with sickle cell disease, particularly boys.
  - Recurrent haematuria is due to medullary infarction, which leads to renal papillary necrosis. Sometimes, sloughing of the papilla may lead to ureteric obstruction and renal failure.
  - Renal medullary carcinoma occurs almost exclusively in people with the sickle cell trait, sickle cell disease, and occasionally sickle cell anaemia and may present with haematuria. It is aggressive and has usually metastasized by the time the symptoms (weight loss, haematuria, abdominal pain or back pain) have occurred.

What are the complications of sickle cell disease during pregnancy?

- **Sickle cell disease is associated with maternal and fetal complications, including:**
  - Perinatal mortality.
  - Premature labour.
  - Fetal growth restriction.
  - Acute painful crises.
- **Some studies have described increased risks of:**
  - Spontaneous miscarriage.
  - Stillbirth.
  - Antenatal hospitilization.
  - Maternal mortality.
  - Delivery by Caesarean section.
  - Infection — particularly urinary tract infection.
  - Thromboembolic events.
  - Antepartum haemorrhage.
  - Pre-eclampsia and pregnancy-induced hypertension.

Diagnosis:

When to Suspect sickle cell disease?

- **Most cases of sickle cell disease are diagnosed as part of the National Newborn Screening Programme in the UK** — in children who have not undergone screening (for example, if they have travelled from overseas), cases present with symptoms in infancy or early childhood.
- **Infants do not usually manifest any signs and symptoms until they are 6 months old, however very young infants may present with signs and symptoms of:**
  - Haemolysis (jaundice, pallor, or tachycardia).
  - Splenic sequestration crisis (pallor, tachycardia, or shock).
- **In children aged 4 months and older, signs and symptoms of sickle cell disease include:**
  - Swelling of the joints, especially dactylitis (painful swelling of the hands or feet as a result of vaso-occlusion) — dactylitis is a common presenting symptom in infants aged 9–18 months, but many children do not experience this and may only present later with vaso-occlusion affecting the long bones.
    - This affects 30% of children in the first year of life and usually resolves in a few days.
  - Leukocytosis in the absence of infection.
  - Invasive infection.
  - Protuberant abdomen (due to an enlarged spleen), often with umbilical hernia.
  - Cardiac systolic flow murmur secondary to anaemia.
  - Maxillary hypertrophy with overbite, due to extramedullary haematopoiesis, may also occur.
- **Also suspect sickle cell disease if a person is in a** high-risk group **and presents with features of:**
  - An acute crisis, or with a history of features consistent with an acute crisis.
  - A chronic complications of sickle cell disease.

When to suspect an acute sickle cell crisis?

Suspect an acute sickle cell crisis in a person with sickle cell disease who presents with a sudden onset of pain, infection, anaemia, or other symptoms, such as a stroke or priapism.

- There may be a history of a previous crisis (or rarely, sickle cell disease may be undiagnosed).

Assessment of acute sickle cell crisis:

If an acute sickle cell crisis is suspected:

- **Take a history.** Ask about clinical features of the acute complications of sickle cell disease.
- **Examine the person.**
  - Examine for painful, swollen joints in addition to the presence of localized warmth and any tenderness.
  - Check the person's temperature.
  - Check the person's skin for any signs of pallor.

        - Acute splenic sequestration may present with pallor, shock, tachycardia, and lethargy. Because it is most common in young children, parents are taught to palpate the spleen, and they may report an acute increase in splenic size.
        - Transient red cell aplasia is characterized by pallor (due to anaemia, which can be severe), tachypnoea and tachycardia without splenomegaly and is most commonly due to parvovirus infection. Associated symptoms may include fever, headache, myalgia, arthralgia, and respiratory and gastrointestinal symptoms.
        - Jaundice — may be present with acute abdominal pain in people with gallstones.
    - Palpate for splenomegaly.
- **Investigations that may be appropriate, depending on signs and symptoms include:**
    - Full blood count with differential and reticulocyte count.
    - Renal and liver function tests.
    - Cultures of blood, urine, and any other potential site of infection.
    - X-ray.
    - CT scan.
    - MRI scan.

Management:

Screening:
From birth onwards.
What is the national screening programme for sickle cell disorders?

- **The Sickle Cell and Thalassaemia (SCT) Screening Programme screens for:**
    - Genetic carriers for sickle cell, thalassaemia, and other haemoglobin disorders.
    - Sickle cell disease.
    - Thalassaemia.
    - Haemoglobin disorders.
- **The British Society for Haematology advises that:**
    - The following people should also be screened:
        - People from high-risk groups who are about to undergo an operation or receive anaesthesia.
        - Women who are having assisted conception.
        - Women who are being investigated for infertility.
        - Women in high-risk groups for haemoglobinopathies before conception — if an abnormality is found, then partners should be offered testing.
    - Pre-marriage screening for haemoglobinopathies is not usual in the UK, but for some religious or ethnic groups, pre-marital screening for beta-thalassaemia heterozygosity may be more acceptable than pre-conceptual or antenatal screening.

Management of sickle cell crisis:
From birth onwards.
How should I manage a person having a sickle cell crisis?

- **Conduct a clinical assessment in people with sickle cell disease who present with signs or symptoms of acute sickle cell crisis.**
- **Urgently refer people with any of the following signs or symptoms:**
    - Severe pain not controlled by simple analgesia or low dose opioids.
        - All people presenting with an acute painful sickle cell episode should be offered analgesia within 30 minutes of presentation in secondary care.
    - Dehydration caused by severe vomiting or diarrhoea.
    - Severe sepsis.
        - Temperature over 38.5°C, or over 38°C in children aged under 2 years, temperature less than 36°C, or hypotension.
    - Symptoms or signs of acute chest syndrome including tachypnoea, oxygen saturation more than 5% below steady state, signs of lung consolidation.
    - New neurological symptoms or signs.
    - Symptoms or signs of acute fall in haemoglobin.
    - Acute enlargement of spleen or liver over 24 hours, particularly in young children.
    - Marked increase in jaundice.
    - Any change in vision.
    - Haematuria.
    - Fulminant priapism lasting more than 2 hours or worsening of recurrent episodes.
- **For all people with fever (temperature of 38°C or higher) who do not require referral:**
    - Offer empirical broad-spectrum antibiotic treatment with choice guided by local microbiological guidelines or haemoglobinopathy teams.
- **Consider empirical broad-spectrum antibiotic treatment in people who do not have a fever, but have signs and symptoms of infection.**
- **For people with mild to moderate pain that can be managed at home.**
    - The person and/or their family/carers will often have received an individual care plan from secondary care with clear guidance on how to manage their pain, and when to seek medical advice.
    - If they do not have a care plan, prescribe paracetamol and/or ibuprofen (avoid ibuprofen if the person has renal impairment or significant proteinuria). Weak opioids can be used for more severe pain (dihydrocodeine in children aged under 13 years and codeine phosphate in people aged over 13 years.
        - Arrange an urgent assessment in hospital if the pain is not controlled by these measures.
    - Offer advice about:
        - Coping strategies including distraction techniques like games, computers, and television.
        - Increasing fluid intake as dehydration may prolong the painful episode.
        - Avoiding triggers, such as exposure to cold or windy weather, excessive physical activity and dehydration.
- **Advise parents/carers to seek urgent medical advice if a child has any of the following signs or symptoms:**

- Fever.
- Respiratory symptoms or other signs of infection.
- Priapism.
- Unusual pallor.
- Weakness (without pain), tingling, loss of speech, or any neurological complications.

Management of chronic complications:

From birth onwards.
Overview of management

The management of chronic complications of sickle cell disease needs a multi-disciplinary approach. Children and adults are seen regularly in secondary care. However, it is recommended that as much of the management as possible should take place in the community.

- **Primary care clinicians should:**
  - Admit or refer people immediately if they have new symptoms of cerebrovascular disease, as acute strokes are devastating in both children and adults.
  - Manage chronic pain with care.
  - Treat chest infections early, especially in people with chronic sickle lung.
  - Refer people with eye problems urgently.
  - Refer people for consideration for cholecystectomy if they have symptomatic gallstones.
  - Address concerns about impaired growth or delayed puberty.
  - Manage leg ulcers appropriately.
  - Refer people with suspected pulmonary hypertension, or worsening symptoms.
  - Manage minor episodes of priapism. It is important to enquire about this issue as the person may be too embarrassed to report it.
  - Manage nocturnal enuresis, which is common because of the production of large quantities of dilute urine.
  - Identify renal problems early, and treat hypertension aggressively.
  - Refer people with sleep apnea.
  - Advise on measures to prevent (or reduce the risk of) complications.

How should I manage cerebrovascular disease in a person with sickle cell disease?

People who have had a stroke will be managed in secondary care with regular follow up. Most people will have regular blood transfusions (preferably exchange transfusions) to keep their sickle haemoglobin percentage below 30% indefinitely. This means that people are also monitored for iron overload. All children should have annual transcranial Doppler imaging from 2 years of age.

How should I manage chronic pain in a person with sickle cell disease?

All types of chronic pain in sickle cell disease require multi-disciplinary support, with the general approach being similar to other types of chronic pain.

How should I manage chronic sickle lung in a person with sickle cell disease?

Children and adults with chronic sickle lung disease will be managed in secondary care.

- **Management in primary care should include:**
  - Treating any chest infection early and referring when appropriate.
  - Giving advice on smoking cessation, where necessary.
  - Ensuring vaccinations are up to date.

How should I manage eye problems in a person with sickle cell disease?

People with sickle cell disease should be evaluated at least every 2–3 years by an ophthalmologist if they have no evidence of retinopathy and remain asymptomatic, and people with retinopathy should be reviewed at least annually. Ocular problems are very rare in early childhood and are found most commonly in people aged 15–30 years.

- **Refer people immediately if they present with an acute change in vision** — this may be due to retinal artery occlusion, vitreous haemorrhage, and retinal detachment.

How should I manage gallstones in a person with sickle cell disease?

Gallstones are very common and often cause intermittent abdominal pain.

How should I manage impaired growth in a child or young person with sickle cell disease?

Growth (weight and height) will usually be monitored in secondary care as part of an annual review.

- **Consider:**
  - Zinc supplementation if growth is retarded (seek specialist advice).
  - Referral to a paediatric dietitian if the child is hospitalized frequently.
  - Referral to a paediatric endocrinologist if there are no physical signs of puberty by 14 years of age in a girl, and by 14 years 6 months of age in a boy.
- Reassure children with delayed growth and delayed puberty (and/or their parents/carers) that they will reach a normal adult height.

How should I manage leg ulcers in a person with sickle cell disease?

Leg ulcers will often be managed in secondary care by a multi-disciplinary team. However, some aspects of care may be undertaken in primary care depending on the care plan and local guidelines.

- **Primary care management should include:**
  - **Prescribing adequate pain relief**
  - **Suspecting underlying osteomyelitis if the person has:**

        - Fever.
        - Localized warmth, tenderness, swelling.
        - A high leukocyte-count.
        - A high C-reactive protein level.
    - **Seeking specialist advice if** the ulcer fails to respond to conservative measures.

How should I manage episodes of priapism in a person with sickle cell disease?

Priapism is common and can be an acute or chronic complication of sickle cell disease. It is important to enquire about this issue as the person may be too embarrassed to report it.Patients should be managed in conjunction with a urologist with a specialist interest in sickle cell disease-related priapism.

- **Provide men with advice on conservative management of priapism:**
    - Gentle exercise, such as jogging.
    - Warm baths.
    - Analgesia.
    - Trying to urinate.
    - Keeping warm.
    - Keeping hydrated at bedtime.
- **Advise the man that if priapism persists for more than 1 hour he should attend hospital for initial management, which may include:**
    - Pain relief.
    - Hydration.
    - Oxygenation.
    - An alpha-adrenergic agent (for example, etilefrine).
- First-line preventative therapy is alpha-adrenergic agonists and anti-androgens.

How should I manage nocturnal enuresis in a child with sickle cell disease?

Nocturnal enuresis will often be managed in secondary care, but supportive care and advice are also provided in primary care.

- **Explain** to the child and/or the parents/carers that enuresis is common in people with sickle cell disease because they produce large quantities of dilute urine, and that most cases resolve spontaneously.
- **Give general advice on the management of enuresis.**
    - Be aware that children with sickle cell disease tend not to respond to behavioural management techniques, such as star charts and mattress alarms.
- **Consider treatment with oral or nasal desmopressin in children who do not respond to routine advice and management —** this is a useful adjunct particularly for school trips or sleepovers.
- **Refer children to:**

- An ear, nose, and throat specialist if the history is suggestive of obstructive sleep apnoea, as hypoxia may exacerbate nocturnal enuresis.
- A specialist enuresis clinic if the child is still enuretic by 7 years of age.

How should I manage pulmonary hypertension in a person with sickle cell disease?

People with pulmonary hypertension will be managed in secondary care. Pulmonary hypertension is often asymptomatic (even when severe) and symptoms may be non-specific.

How should I manage renal disease and hypertension in a person with sickle cell disease?

People with sickle cell should be monitored at least annually for symptoms or signs of renal disease (for example, urinary tract infection, haematuria), for hypertension, and for the presence/progression of albuminuria, proteinuria and declining renal function.

- **Primary care clinicians will be involved in monitoring and treating hypertension and monitoring for chronic kidney disease (CKD).**
- **People with acute renal failure or with evidence of declining renal function should be managed jointly with a renal physician.**
- **In adults, aim for a target blood pressure of under:**
  - 130/80 mmHg in people who have albuminuria (albumin creatinine ratio [ACR] of 3.5 mg/mmol or more).
    - If blood pressure is 130/80 mmHg or higher offer an angiotensin-converting enzyme (ACE) inhibitor, angiotensin-II receptor blocker (ARB), or a calcium channel blocker.
  - 140/90 mmHg in people with no albuminuria.
    - If blood pressure is 140/90 mmHg or higher offer a calcium channel blocker.
- **Advise a good fluid intake (a minimum of 3 litres per day) to prevent dehydration,** as people with sickle cell disease produce large quantities of dilute urine.
- **Refer children with hypertension for further investigation.**
- **Refer people urgently if they present with symptoms suggestive of renal medullary cancer — haematuria, abdominal or back pain, and weight loss.**
  - New-onset haematuria should be investigated, regardless of age, to exclude malignancy.
  - Refer using the suspected cancer pathway referral (for an appointment within 2 weeks).

How should I manage sleep apnoea in a person with sickle cell disease?

Sleep apnoea is common in children and adults with sickle cell disease and may be due to tonsillar hypertrophy or other causes of sleep disordered breathing.

- Refer children and adults with a history suggestive of sleep apnoea for sleep studies.

Prevention of complications:
From birth onwards.
What are the recommended immunizations for people with sickle cell disease?

- All people with sickle cell disease should be fully vaccinated according to the national immunization schedule.

- **For children with sickle cell disease, ensure that they receive:**
    - Immunizations in accordance with the Childhood Immunization Programme.
    - Pneumococcal polysaccharide vaccine (PPV23) at 2 years.
    - Influenza vaccine annually from the age of 6 months.
    - Meningitis ACWY vaccine — the schedule depends on the age the child is first diagnosed or present
- **Older children and adults diagnosed with sickle cell disease, regardless of previous vaccination, should receive:**
    - PPV23 vaccine every 5 years.
    - One dose of MenB and MenACWY, followed by an additional MenB vaccine dose 4 weeks later.
    - Influenza vaccine annually.
- **Other vaccines that may be appropriate include:**
    - Hepatitis B vaccine if they have not received it, or have previously received it but are non-immune.

What antibiotic treatment should I prescribe to protect against infection in a person with sickle cell disease?

- **Penicillin prophylaxis should be offered to all children with sickle cell disease, started by 3 months of age and continued until the child is 5 years old.**
- **If the person is allergic to penicillin, oral erythromycin should be given.** The recommended doses are:
    - 1 month to 2 years — 125 mg twice a day.
    - 2–8 years — 250 mg twice a day.
    - Over 8 years — 500 mg twice a day.
- **Lifelong prophylactic antibiotics should be offered to people considered at continued high risk of pneumococcal infection.**
    - Prescribe antibiotic prophylaxis for adults who prefer to continue.

How should I prescribe folic acid for a person with sickle cell disease?

- Folic acid supplementation is unlikely to be necessary on a regular basis in adults with sickle cell disease with a balanced diet.
- Consider prescribing folic acid for people with a diet that does not contain adequate folic acid, or in whom there is proven folate deficiency.
    - Seek specialist advice before prescribing folic acid to determine the optimum dose.
- A dose of 5 mg folic acid once a day is recommended for pregnant women and for women trying to conceive.

What should I advise on malaria prophylaxis for people with sickle cell disease?

- **Explain that** sickle cell disease does not protect against malaria and that malaria is likely to be severe because of splenic hypofunction.

- **Advise that** all adults and children travelling to an area where malaria is endemic must take adequate prophylaxis, and adherence is very important.

What interventions are available in secondary care to prevent chronic complications in people with sickle cell disease?

- **The detailed management of sickle cell disease in secondary care is beyond the scope of this guidance.**
- Parents or carers will receive comprehensive advice on caring for children with sickle cell disease which should be reinforced in primary care if necessary, including:
  - When to seek urgent medical assistance.
  - How to examine their child for pallor and an enlarging spleen.
  - The importance of keeping warm, avoiding sudden changes in temperature, and avoiding dehydration (by maintaining a good fluid intake).
  - The importance of adherence to prophylactic antibiotics and the immunization schedule.
  - When to seek early medical advice, for example if the child has a fever, respiratory symptoms, signs of infection, unusual pallor, an enlarged (or enlarging) spleen, signs of stroke (weakness, tingling, or loss of speech), or painful crisis and dactylitis.
  - To report any concerns, for example about a child's development or any deterioration in school achievement.
  - To seek advice about intended travel to a foreign country, so that travel immunization and malarial prophylaxis are provided.
- **Management in secondary care may include:**
  - Regular psychological support for children — cognitive behavioural therapy may also be offered to children experiencing frequent pain episodes and emotional difficulties.
  - Elective blood transfusion — for example, for people with pulmonary hypertension, acute priapism, or severe anaemia.
  - Hydroxycarbamide(hydroxyurea) treatment — for example, for people who have had recurrent hospital admission for acute chest syndrome or acute painful crises.
  - Surgery — for example, to relieve pain due to avascular necrosis, or splenectomy for people who have had two or more episodes of acute splenic sequestration.
  - Stem cell transplantation — this is the only cure for sickle cell disease and may be considered in children and some adults with severe sickle cell disease.

Management of Sickle cell trait:

From birth onwards.
How should I manage people with sickle cell trait?

As sickle cell trait is usually asymptomatic, management consists mainly of providing appropriate advice.

- **Explain to people with sickle cell traits and/or their family/carers that:**
  - They should very rarely have symptoms. However, they are at risk of a vaso-occlusive episode if they become oxygen deprived. They should therefore:
    - Be careful at high altitudes (for example, long haul flights, mountain climbing).

        - Inform the anaesthetist that they are sickle cell carriers, if they are going to have an anaesthetic.
    - Other known triggers are high atmospheric pressure (such as scuba diving) and exercise — advise them to stay hydrated.
    - They have a 1 in 2 chance of passing the sickle haemoglobin gene to their child. If the other parent is also a carrier, there is a 1 in 4 chance that their child will have sickle cell disease.
    - It is important to have malaria prophylaxis if they are visiting an area where malaria is endemic.
- **Refer children and adults with haematuria.**
- **Refer children and adults urgently if they present with symptoms suggestive of** renal medullary cancer **— haematuria, weight loss, loin pain, fever, and abdominal pain.**

Contraception:

From age 13 years onwards.
Which contraception is suitable for women with sickle cell disease?

When advising on contraception, consider the potential harms in the context of the benefits (that is, preventing potentially serious complications **caused by pregnancy in a woman with sickle cell disease).**

- **In general, women with sickle cell disease without complications can use any method of contraception. Options include:**
    - Combined hormonal contraceptives (pill, patch, or vaginal ring).
    - Progestogen-only contraception (pill, implant, or injectable).
    - Intrauterine contraception (levonorgestrel intrauterine system [LNG-IUS] or copper intrauterine devices [Cu-IUD]).
    - Surgical sterilization.
- **Advise the woman that:**
    - Sole use of barrier methods or behaviour-based methods of contraception may not be an appropriate choice for them due to higher typical-use failure rates of these methods of contraception.
    - Long-acting reversible contraception (such as Cu-IUD, LNG-IUS, progestogen-only injectables, and the progestogen-only implant) are more effective at preventing unwanted pregnancy than user-dependent methods.
- **Be aware that:**
    - Combined hormonal contraceptives are contraindicated in women with pulmonary hypertension.
    - There may be an increased risk of menorrhagia with the Cu-IUD.
    - Surgical sterilization in women is associated with increased risk of pulmonary, cardiac, and neurological complications, and a possible increased risk of wound infections. Women with sickle cell disease who desire sterilization should receive appropriate counselling.
    - In men with sickle cell disease, there is no medical reason to deny vasectomy.

TOPIC: Beta thalassemias

The thalassaemias are a group of genetic disorders characterised by a reduced production rate of either alpha or beta chains
There are two beta globin genes (one inherited from each parent). Beta thalassemia is caused by reduced production of beta chains and accumulation of excess alpha chains. This may be due to variants that reduce the expression of beta globin ($beta^+$) or completely eliminate beta globin expression ($beta^0$). The severity of disease correlates with the amount of normal beta globin production. Terminology has shifted from using the "major, intermedia, minor" designation to referring to the disorder as transfusion-dependent or non-transfusion-dependent.

1. **Normal**: genotype β_2/β_2.
2. **β-thalassaemia trait** (genotype $-/\beta_2$): HbA_2 >4%. Slightly anaemic, low MCV and MCH; clinically asymptomatic.
3. **β thalassaemia intermedia** (genotype -/βo or β+/β+): high HbF, variable. Anaemic (symptoms usually develop when the haemoglobin level remains below 7.0 g/dL), very low MCV and MCH; splenomegaly, variable bone changes, variable transfusion dependency.
4. **β thalassaemia major** (genotype -o/-o): HbF >90% (untransfused). Severe haemolytic anaemia, very low MCV and MCH; hepatosplenomegaly, chronic transfusion dependency.
    a. Presentation of β thalassaemia major in infancy often includes failure to thrive, vomiting feeds, sleepiness, stunted growth and irritability.

Beta-thalassaemia trait
Beta-thalassaemia trait is an autosomal recessive condition characterised by a mild hypochromic, microcytic anaemia. It is usually asymptomatic

Features

- mild hypochromic, microcytic anaemia - microcytosis is characteristically disproportionate to the anaemia
- HbA2 raised (> 3.5%)

Investigations:

- In beta-thalassemia trait, FBC will show a microcytic anaemia, with:
    - Normal to moderately low Hb
    - Low MCH, usually 19-23 pg
    - Low MCV, usually 60-70 fL
- Haemoglobin electrophoresis with hemoglobin F and A2 quantitation is used to make a diagnosis of beta-thalassemia trait. Levels of haemoglobin alpha 2 gene (HbA2) are higher in beta-thalassemia carriers.

Unlike iron deficiency anemia, the RBC count in thalassemia may be normal or increased. Iron stores are also normal or increased. The RDW is normal in some patients with thalassemia minor. In others, it may be elevated, but the elevation is not usually to the levels seen in iron deficiency anemia
DDx:

- **Iron deficiency anaemia:**
    - Similarities: microcytic anaemia (low Hb, low Hct)
    - Differences: acquired condition, due to lack of iron. No structural abnormality. Normal levels of HbA2
- **Beta-thalassemia major:**

Similarities: same genetic mutation on HBB gene on chromosome 11
Differences: homozygous for mutation. More likely to be symptomatic, requiring medical intervention. Shortened life expectancy

- **Sickle cell disease**

Management

- The majority of patients with beta-thalassemia trait do not require any treatment.
  If there are concerns about symptomatic anaemia, ferritin levels should be checked to assess whether iron supplementation is required or safe.
    - If ferritin levels are low, start iron supplementation
    - If ferritin levels are normal, patients should be advised not to take iron supplements due to the risk of iron overload and the lack of therapeutic benefit
    - We suggest folic acid supplementation if there is evidence of ongoing hemolysis (eg, 1 to 2 mg daily), and we avoid iron supplementation unless there is concomitant iron deficiency.

TOPIC: Aplastic anemia

Characterised by pancytopenia and a hypoplastic bone marrow
Aplastic anemia is a bone marrow failure disorder characterized by pancytopenia with bone marrow hypoplasia/aplasia.
Peak incidence of acquired = 30 years old

Features:

- Normochromic, normocytic anemia
- Leukopenia, with lymphocytes relatively spared
- Thrombocytopenia
- May be the presenting feature acute lymphoblastic or myeloid leukaemia
- A minority of patients later develop paroxysmal nocturnal haemoglobinuria or myelodysplasia

Causes:

- Idiopathic (most common)
- Congenital: Fanconi anemia, dyskeratosis congenita
- Drugs: cytotoxics, chloramphenicol, sulphonamides, phenytoin, gold, NSAIDs
- Toxinc chemicals: benzene
- Infections: parvovirus, hepatitis, EBV, HIV
- Immune disorders : SLE
- Radiation
- Others: pregnancy, thymoma

History:

- Exposure to drugs/chemicals, viral infection.
- Family history of cytopenias and/or somatic findings suggestive of an inherited disorder.

Examination:

- Inspect for pallor petechiae, mucosal ulcers, and signs of infection or bleeding
- Palpate liver, spleen and lymph nodes

Investigations:

- CBC
- Blood smear
- RFT
- LFT
- Vit B12
- Folate
- Bone marrow aspiration and biopsy

Management:

The course of non-severe aplastic anemia is variable. The majority of cases have a mild course requiring no treatment.
Treatment options include:

- Removal of any underlying cause - eg, drug-induced aplastic anemia.
- Haemopoietic stem cell transplant (HSCT)
- Immunosuppressive therapy

TOPIC: Platelets - abnormal count and cancer

Definition:

- Platelets, or thrombocytes, are anucleate haematopoietic cells derived from bone marrow megakaryocytes — they circulate for 7–10 days before being cleared by the cells of the reticuloendothelial system.
- They have the following roles:
  - Haemostasis and thrombosis — platelets express and secrete adhesion molecules that accumulate in response to injury. These adhesion molecules promote agglomeration of leucocytes and granulocytes.
  - Inflammation, infection, and immune host response — platelets secrete immune modulators. These are chemotactic for neutrophils, monocytes, and lymphocytes. This interaction results in the formation of platelet-granulocyte or platelet-leukocyte aggregates that trigger a further inflammatory response. Platelets also engulf microbes and prevent dissemination of bacteria by clot formation.
- Platelet count varies with age, sex, and ethnicity, and there is also a genetic component (for example, familial thrombocytosis).
- A normal platelet count usually lies within the range 150–450 × 109/L.
- **Thrombocytosis** is defined as a platelet count greater than 450 × 109/L.
- **Thrombocytopenia** is defined as a platelet count below 150 × 109/L.

Causes of thrombocytosis:

- There are a wide range of causes of a raised platelet count — they can be classified as primary or secondary.
- **Primary thrombocytosis** is caused by an abnormality in bone marrow function that arises from an expansion of a mutated haematopoietic stem cell or myeloid progenitor cells that give rise to megakaryocytes. It can be classified into the following diseases:
  - Polycythaemia rubra vera.
  - Chronic myeloid leukaemia.
  - Essential thrombocythaemia.
  - Primary myelofibrosis.
  - Unspecified myeloproliferative disorders.
- **Secondary (or reactive) thrombocytosis** accounts for over 80% of all recognized cases of thrombocytosis. It is driven by thrombopoietic growth factors released in response to an inflammatory stimulus and release of inflammatory cytokines caused by a variety of conditions including:
  - Chronic inflammation (for example, rheumatoid arthritis or systemic lupus erythematosus).
  - Drug therapy (for example, corticosteroids).
  - Haemolysis.
  - Iron deficiency — this is the most common cause of secondary thrombocytosis.
  - Severe systemic infections (such as pericarditis or endocarditis).
  - Some specific cancers.
  - Splenectomy, hyposplenism.

    - Surgery, bleeding, and other trauma.
- The thrombocytosis associated with certain cancers is mediated by a sequential tumour cell-induced platelet aggregation (TCIPA) process firstly involving the release of chemical mediators that sets in train a physical process involving the following steps:
    - Stabilization and protection of cancer cells in vascular tissue.
    - Stimulation of cancer cell proliferation.
    - Promotion of cancer cells extravasation through blood vessels.
    - Enhancing cancer cell interaction.
- Cancer cells then trigger further coagulation cascade by generating thrombin, which in turn induces platelet activation, thus creating a positive feedback loop of further metastases.

Causes of thrombocytopenia:

- **The causes of thrombocytopenia are diverse, but can be considered as arising from three main causes:**
    - Reduced platelet production as a result of marrow-based disorders — for example, malignancy, myelofibrosis, sarcoidosis, nutritional deficiency, congenital disorders.
    - Increased platelet destruction — for example, immune thrombocytopenic purpura, disseminated intravascular coagulation, thrombotic thrombocytopenic purpura, autoimmune thrombocytopenia, and drug-related thrombocytopenia (for example, quinidine, quinine, nonsteroidal anti-inflammatory drugs [NSAIDs], penicillin, and anticonvulsants).
    - Increased platelet sequestration — for example splenomegaly, due to portal hypertension that may be caused by cardiac failure, hepatic vein thrombosis or vena cava thrombosis (Budd-Chiari syndrome), cirrhosis (such as due to chronic viral hepatitis or alcoholic liver disease) and, rarely, arteriovenous malformation of the splenic vessels.
- **Dilutional** — for example, after blood transfusion (this may occur after 1 or 2 units of blood) or pregnancy.
- **Cancer-related thrombocytopenia** — this is often caused by myelosuppression due to bone marrow infiltration or myelodysplasia (for example, from acute leukaemias).
    - It can also be caused by specific cancers, including small cell lung cancer, and more rarely, paraneoplastic syndromes such as idiopathic thrombocytopenic purpura (ITP) as part of an association with some solid tumours such as breast and lung cancer.
    - Treatment-induced thrombocytopenia can be caused by radiation or systemic chemotherapy.
- **Pseudothrombocytopenia** — a spurious laboratory result. Around 1–2% of the population have an ethylene diamine tetra acetic acid (EDTA)-dependent antibody in their circulation, resulting in pseudothrombocytopenia. These individuals have no history of bleeding and have normal circulating counts.
    - This can be confirmed by reviewing the blood film.

Diagnosis:

Assessment of thrombocytosis:

- **Review any previous results for similarly raised counts or a rising trend in platelets.**
- **Take a history and ask about:**

    - Recent infection, chronic inflammatory conditions, surgery, cancer, prior splenectomy, iron deficiency anaemia.
    - Family history of thrombophilia.
    - Family history of cancer.
    - Any symptoms of possible cancer diagnoses, including:
        - Change in bowel habit.
        - Dysphagia.
        - Dyspnoea.
        - Fatigue.
        - Persistent dyspepsia in people aged over 55 years.
        - Post-menopausal bleeding in women aged over 55 years.
        - Rectal bleeding.
        - Unexplained cough.
        - Unexplained haemoptysis.
        - Weight loss.
    - Any other symptoms, including vasomotor symptoms, such as headache, lightheadedness, chest pain, paraesthesia, vertigo, dizziness, syncope, seizures, and erythromelalgia (characterized by burning pain and dusky congestion of the extremities).
    - Thrombotic or haemorrhagic complications.
    - Cardiovascular risk factors.
    - History of previous pregnancies if appropriate (for example, miscarriages, bleeding, or fetal loss).
- Perform an examination and look for signs of:
    - Anaemia.
    - Infection.
    - Malignancy.
    - A surgical scar consistent with splenectomy.
    - Livedo reticularis (a purplish mottled discolouration of the skin, usually on the legs) — this is typically described as lacy or net-like in appearance and is seen in several connective tissue diseases such as lupus, antiphospholipid syndrome, or Sneddon's syndrome.
- Consider other secondary causes of raised platelets, such as infection or inflammation, and arrange necessary investigations directed by the suspected diagnosis from clinical evaluation.
    - Review the results of the blood film — which may show other features to indicate an underlying cause, including acute infective, or inflammatory processes.
        - In reactive thrombocytosis the platelets are mostly smaller with a low mean platelet volume compared with essential thrombocytosis or people with normal platelets. The blood film may show other features to indicate an underlying cause, including acute infective, or inflammatory processes.
    - Check for iron deficiency

    - Measure C-reactive protein (CRP) and erythrocyte sedimentation rate (ESR) levels — these are usually, but not always elevated in reactive thrombocytosis.
- If appropriate, depending on the clinical evaluation, consider organizing:
    - A chest X-ray in people aged 40 years or over.
    - Pelvic ultrasound in women aged 55 years or over.
    - A faecal immunohistochemical test (FIT).
    - A Janus kinase-2 (JAK-2) test, JAK-2 is a genetic mutation that may be present in people with essential thrombocythaemia and can indicate a diagnosis of polycythaemia rubra vera.
- Consider specialist referral for colonoscopy or other assessments in secondary care, depending on local referral pathways.
    - Some areas have a route for referral for suspected cancers of indeterminate origin.

Assessment of thrombocytopenia:

- **Review any previous results to look for similar lower counts or a falling trend in platelets.**
- **Take a history and ask about:**
    - Alcohol intake, dietary habits, prescribed medicines (for example, quinine, heparin), over-the-counter medicines, recent vaccinations, and recreational drugs.
    - Recent infections, rheumatological conditions, malignancy, pregnancy, recent surgery or transfusion.
    - Recent travel (for example to malaria-endemic areas).
    - Family history of thrombocytopenia.
    - Symptoms of possible diagnoses, for example:
        - Bone pain.
        - Change in bowel habit.
        - Chills, night sweats, fever, rigors.
        - Diarrhoea, nausea, vomiting.
        - Epistaxis.
        - Excessive thirst, urination, and constipation.
        - Fatigue.
        - Gastrointestinal bleeding.
        - Genitourinary bleeding.
        - Gum bleeding.
        - Heavy menstrual bleeding.
        - Night sweats.
        - Post-menopausal bleeding.
        - Skin bruising/bleeding.
        - Weight loss.
- **Perform a physical examination and look for signs of an underlying cause, including:**
    - Bleeding at the back of the buccal mucosa advancing to the front.

    - Bruising or rashes.
    - Digital ischaemia and/or swollen extremities.
    - Hair loss, bradycardia, enlarged thyroid gland.
    - Jaundice and hepatomegaly.
    - Joint deformity.
    - Pallor.
    - Petechiae or ecchymoses.
    - Retinal haemorrhage, retinal changes.
    - Skeletal abnormalities.
    - Spider telangiectasis on the chest, palmar erythema, ascites, pedal oedema, spider angiomas.
    - Splenomegaly or lymphadenopathy.
    - White spots on the tongue.
- **Arrange necessary investigations directed by the suspected diagnosis from clinical evaluation, including:**
    - Review the blood film.
    - Check full blood count and reticulocyte count.
    - Check prothrombin time.
    - Measure renal and liver function.
- **Other investigations that may be appropriate, depending on the suspected diagnosis, include tests for:**
    - Antiphospholipid antibodies.
    - Antinuclear antibodies.
    - Rheumatoid factor.
    - Thyroid-stimulating hormone.
    - HIV, hepatitis C, and other infectious causes (such as Epstein–Barr virus or varicella).
    - Vitamin B and folate levels.
    - Lactate dehydrogenase.
- Pregnant women beyond 20 weeks' gestation with thrombocytopenia or signs and symptoms such as headache, visual disturbances, right upper quadrant abdominal pain, or elevated blood pressure should be evaluated for preeclampsia and HELLP (haemolysis, elevated liver enzymes, and low platelet count) syndrome.
    - Note: women who are pregnant can also present with other causes of thrombocytopenia that can mimic HELLP syndrome, including disseminated intravascular coagulation (as seen with sepsis and placental abruption) and thrombotic thrombocytopenic purpura.

Interpreting platelet results:

- The normal laboratory reference range for platelets is 150–450 × 109/L.
    - Thrombocytosis is a platelet count greater than 450 × 109/L.
    - Thrombocytopenia is a platelet count below 150 × 109/L — this can be further subdivided into:
        - Mild: over 100 × 109/L.

        - Moderate: 50–100 × 109/L.
        - Severe: less than 50 × 109/L.
- However, it is possible that a person could have a high or a low platelet count as part of their normal haematology.
    - Platelet counts in pregnant women gradually decrease as pregnancy progresses, so the definition of thrombocytopenia will therefore vary.

Cancer causes:

- **Primary causes of thrombocytosis include:**
    - Essential thrombocythaemia — diagnosis requires a sustained thrombocytosis and the exclusion of reactive causes.
    - Polycythaemia vera — this is characterized by an elevated haemoglobin, haematocrit, and red cell mass (although up to 15% of people may present with a marked thrombocytosis and clinical features such as pruritis).
    - Primary myelofibrosis — this can present with a marked thrombocytosis, with or without anaemia, teardrop poikilocytosis, leucocytosis and a leuco-erythroblastic blood film. Clinical features are similar to essential thrombocythaemia.
    - Chronic myeloid leukaemia — the initial clinical manifestation can be isolated thrombocytosis. Other features include an elevated white blood cell (WBC) count (over 100 × 109/L in half of people). Peripheral smear shows WBCs are mostly neutrophils, although basophil and eosinophil counts are also commonly raised.
    - Myelodysplastic syndromes — full blood count shows normochromic or macrocytic red blood corpuscles (RBCs) and morphological abnormalities such as oval macrocytic RBCs and granulocytes with the pseudo-Pelger-Huet anomaly (hypogranular and hypolobulated granulocytes).
    - Unclassified myeloproliferative neoplasms.
- **Thrombocytosis can be a marker for potential cancer, including:**
    - Lung cancer
    - Endometrial cancer
    - Gastric cancer
    - Oesophageal cancer
    - Colorectal cancer.

Non-cancer causes:

- **Non-cancer conditions associated with thrombocytosis include:**
    - Ankylosing spondylitis
    - Chronic hepatitis B.
    - Chronic obstructive pulmonary disease (COPD).
    - Coeliac disease.
    - Functional hyposplenism — people who have reduced splenic function may have reactive thrombocytosis and will have nuclear fragments (Howell-Jolly antibodies) in their red blood cells.
    - Giant cell arteritis.
    - Granulomatosis with polyangiitis.

- Infection — this is typically bacterial in adults.
- Inflammatory bowel disease.
- Iron-deficiency anaemia — iron supplementation generally corrects the platelet count.
- Ischaemic heart disease.
- Raynaud's phenomenon.
- Rheumatoid arthritis.
- Sarcoidosis.
- Segmental colitis associated with diverticulosis
- Thromboembolic disease.
- Trauma, major surgery, or blood loss.

- **Non-cancer causes of thrombocytopenia include:**
  - Medication (for example, antihypertensives, antibiotics, heparin) — thrombocytopenia is usually seen after 2–3 days if the medicine has been used before, or after 1–3 weeks for a new medicine. People with drug-induced thrombocytopenia typically have moderate to severe thrombocytopenia.
  - Nutritional deficiency — folate and B12 deficiency can result in severe thrombocytopenia. Other haematological abnormalities include megaloblastic anaemia and hypersegmented neutrophils.
  - Systemic infections — viral (for example, HIV), bacterial (for example, leishmaniasis), or fungal infections can result in transient thrombocytopenia, but it is generally mild to moderate and is not usually associated with symptoms of bleeding.
    - Thrombocytopenia associated with HIV is common, occurring in at least 20% of people with symptomatic disease.
  - Alcohol misuse — platelet production is directly suppressed by alcohol, but thrombocytopenia may result from hypersplenism and nutritional deficiency. It may be associated with a megaloblastic anaemia and ringed sideroblasts.
  - Aplastic anaemia — marrow failure and low megakaryocyte numbers result in low platelet levels.
  - Autoimmune disorders — for example, systemic lupus erythematosus (SLE), antiphospholipid syndrome, sarcoidosis.
    - Thrombocytopenia can occur in up to 25% of people with SLE.
  - Chronic liver disease — thrombocytopenia may be seen in up to 76% of people with chronic liver disease.
  - Gestational thrombocytopenia — maternal thrombocytopenia between 100 × 109/L and 149 × 109/L in asymptomatic pregnant women with no history of bleeding is usually due to gestational thrombocytopenia. Onset can occur at any point in pregnancy, but most commonly occurs in the mid-second to third trimester.
  - HIV infection — thrombocytopenia lasting >4 weeks is an HIV indicator condition.
  - Inherited disorders (for example, dyskeratosis congenita, Fanconi anaemia).
  - Paroxysmal nocturnal haemoglobinuria — a clonal stem cell defect leads to maturation defects in megakaryocytes.

- Primary immune thrombocytopenia (ITP) — this is an acquired disorder and is a diagnosis of exclusion. It is characterized by an isolated thrombocytopenia for which no cause can be found.
    - A previous infection has been reported in 55–60% of paediatric cases.
- Thrombotic thrombocytopenic purpura (TTP) — this can be congenital or acquired. Most cases are idiopathic but TTP can be associated with other diseases and conditions, such as pregnancy and postpartum, HIV infection, and some drugs.

Management:

From age 16 years onwards.
How should I manage a person with thrombocytosis?

- **Conduct an assessment to determine the possible underlying cause of thrombocytosis** — as the underlying condition may be identified by this assessment.
- **Consider the potential of a possible diagnosis of cancer, if in the course of reviewing a full blood count a raised platelet count is found.**
- **For people in whom cancer is not suspected with platelet counts:**
    - Over 1000 × 109/L — arrange urgent referral.
    - Within the range 600–1000 × 109/L — arrange urgent referral if in association with:
        - Recent arterial or venous thrombosis (including deep vein thrombosis [DVT], pulmonary embolism [PE], cerebrovascular accident, transient ischaemic attack [TIA], myocardial infarction [MI], unstable angina, peripheral vascular disease [PVD]).
        - Neurological symptoms.
        - Abnormal bleeding.
        - Age over 60 years.
        - Other significantly abnormal full blood count (FBC) indices.
    - Persistent and unexplained over 450 × 109/L (for more than 3 months) or over 600 × 109/L on at least two occasions (4–6 weeks apart), or within the range 450–600 × 109/L with other haematological abnormalities — refer to haematology for further investigation and management.
- **For people who do not require urgent referral:**
    - With a known underlying cause — manage appropriately according to local and national guidelines.
        - The thrombocytosis does not require specific therapy — it usually resolves once the underlying cause is treated.
    - Where no underlying cause is identified, and the person remains asymptomatic — repeat the full blood count in 4–6 weeks.

How should I manage a person with thrombocytopenia?

- **Conduct an assessment to determine the possible underlying cause of thrombocytopenia.**
- **Consider the potential for cancer, if in the course of reviewing a full blood count a lowered platelet count is found.**
- **For people in whom cancer is not suspected with platelet counts:**

- Less than 20 × 109/L, or with active bleeding — arrange immediate referral, or seek urgent specialist advice to arrange direct assessment. The risk of spontaneous haemorrhage increases significantly below 20 × 109/L.
- From 20- 50 × 109/L — arrange urgent referral.
- Within the range 50–100 × 109/L — arrange urgent referral if in association with:
  - Evidence of pancytopenia (haemoglobin less than 100 g/L, neutrophils less than 1 × 109/L).
  - Splenomegaly or lymphadenopathy.
  - Pregnancy.
  - Upcoming surgical or interventional procedures.
- If 50–100 × 109/L who do not meet the other referral criteria — repeat full blood count in 1–2 weeks.
- Less than 100 × 109/L and persistent and unexplained (on at least two occasions 4–6 weeks apart) — refer to haematology, in accordance with local guidelines.

- **For people who do not require referral to haematology:**
  - With a known underlying cause — manage appropriately according to any local and/or national guidelines.
  - Taking a medicine associated with thrombocytopenia — consider stopping the medicine (in consultation with any relevant specialist), review and repeat the count in 1–2 weeks.
    - Platelet counts usually start to recover after 4–5 half-lives of the responsible drug or drug metabolite, so recovery is usually seen within several days to 2 weeks.
  - With a platelet count of 100–150 × 109/L where no underlying cause is identified — repeat the full blood count in 4–6 weeks.
    - Refer people with a lowering platelet count trend, other haematological abnormalities, or if unwell.
    - Close clinical observation may be required in women who are pregnant — platelet counts may decrease before other clinical features of preeclampsia become apparent.
- **For people with thrombocytopenia who are under the care of an oncologist and who have a low platelet count likely to be due to their chemotherapy** — liaise with the specialist and follow any shared care arrangements.

TOPIC: Thrombophilia

Thrombophilia refers to a predisposition to thromboembolism. In practice, the term is used to describe patients who are at significantly increased long-term risk of venous thromboembolism (VTE). Heritable thrombophilia is an inherited tendency for venous thrombosis

Causes :

1. Inherited
    a. Gain of function polymorphisms
        i. factor V Leiden (activated protein C resistance): most common cause of thrombophilia
        ii. prothrombin gene mutation: second most common cause
    b. Deficiencies of naturally occurring anticoagulants
        i. antithrombin III deficiency, protein C deficiency, protein S deficiency

2. Acquired
    a. Antiphospholipid syndrome: Associated with both arterial and venous thrombosis.
    b. Drugs: the combined oral contraceptive pill
    c. Due to liver disease, nephrotic syndrome, disseminated intravascular coagulation or pregnancy.
    d. Myeloproliferative disorders, thrombocytosis or polycythaemia.
    e. Cancer.

Patient usually presents with:

- A strong family history of VTE, VTE at a young age
- VTE which is spontaneous or with minimal provoking factors.
- Thrombosis in an unusual site (eg, mesenteric, portal vein, sagittal sinus thrombosis) or in multiple sites.
- Recurrent VTE, recurrent miscarriage

History:

Take a careful personal and family history, including VTE events and whether there were any provoking factors for the VTE event, such as immobility, surgery, oestrogens, etc.

Counsel females regarding Oral contraceptive pills choice

Management :

- Minimizing VTE risk, ensure mobility and adequate hydration
- Extra precautions and short-term thromboprophylaxis may be needed at times of increased risk - eg, surgery, immobility, pregnancy and postnatally.
- Avoid oestrogen-containing contraceptives and HRT.
- Progestogen-only contraceptives can be used.
- Pre-pregnancy counselling, thromboprophylaxis
- All patients with known thrombophilia or previous VTE - consider short-term thromboprophylaxis at times of increased VTE risk.
- Patients with a first VTE event - long-term anticoagulation is not indicated.
- Patients with ≥2 spontaneous VTEs - consider indefinite anticoagulation.
- Patients with recurrent VTEs linked to a provoking factor (eg, surgery, pregnancy, oestrogen use) may not require long-term anticoagulation but do require prophylaxis during any further high-risk situations.

Investigation:

- CBC and film - looking for myeloproliferative disorders, paroxysmal nocturnal haemoglobinuria, thrombocytosis, polycythaemia.
- Prothrombin time and activated partial thromboplastin time (aPPT).
- Assays for antiphospholipid antibodies, factor V Leiden, prothrombin G20210A, protein C, protein S and antithrombin.
- ESR, CRP, antinuclear antibodies - for connective tissue disorders or inflammation.
- Clotting screen - raised fibrinogen, raised prothrombin, raised factor VIII, plasminogen, factor XII.
- Homocysteine levels.
- Investigations for cardiac disease, liver disease, nephrotic syndrome, or other causes of acquired thrombophilia as appropriate.
- Consider occult malignancy and investigate appropriately.

TOPIC: Immune thrombocytopenia (ITP) in adults

Immune thrombocytopenic purpura (ITP) is an immune-mediated reduction in the platelet count.
Antibodies are directed against the glycoprotein IIb/IIIa or Ib-V-IX complex.
ITP in children most commonly occurs following a viral infection, or occasionally following immunisation.
It is usually a self-limiting disorder which follows a benign course and recovers spontaneously after 6-8 weeks.
In contrast, adults tend to have a more chronic condition and its more common in older females.

Causes:

In ITP, otherwise normal platelets are destroyed, most often in response to an unknown stimulus. This may occur in isolation (primary ITP) or in association with other disorders (secondary).
Causes of secondary ITP include:

- Autoimmune disorders (including antiphospholipid antibody syndrome and SLE).
- Viral infections (including cytomegalovirus, varicella-zoster, hepatitis C and HIV).
- Infection with Helicobacter pylori.
- Medication.
- Lymphoproliferative disorders.

Symptoms:

- may be detected incidentally following routine bloods
- symptomatic patients may present with
  - petechiae, purpura
  - bleeding (e.g. epistaxis)
  - catastrophic bleeding (e.g. intracranial) is not a common presentation

DDX of thrombocytopenia:

- Drug induced thrombocytopenia, e.g. Heparin, quinine, sulfonamides, ibuprofen
- Food and beverages, e.g. quinine containing beverages, walnuts, and certain herbal teas
- Infections, e.g. HIV, Hepatitis C, EBV
- Alcohol
- Rheumatological/autoimmune disorders, e.g. SLE, Rheumatoid arthritis
- Aplastic anemia
- Leukemia

Physical examination:

- Inspect skin and oral mucous membranes for signs of bleeding
- Pallor
- Lymphadenopathy
- Hepatosplenomegaly

Management :

Reassurance and Explanation:

ITP is a disorder of the platelets. Platelets are blood cells that help blood to clot.
People with ITP have a lower than, normal number of platelets because their immune system destroys their platelets. The immune system is the body's infection-fighting system.
Some people with ITP can bruise or bleed much more easily than normal.

Prescription:

- first-line treatment for ITP is oral prednisolone
- pooled normal human immunoglobulin (IVIG) may also be used
  - it raises the platelet count quicker than steroids, therefore may be used if active bleeding or an urgent invasive procedure is required
- splenectomy is now less commonly used.

Referral:

- referral to hematology

Investigation:

- full blood count: isolated thrombocytopenia
- blood film
- a bone marrow examination is no longer used routinely

TOPIC: Bruising

Definition:

- A bruise is a haematoma that forms due to bleeding under intact skin into subcutaneous tissue, caused by vascular damage or injury.
    - A bruise typically initially appears as a red discolouration that does not blanch on pressure. The colour gradually darkens to purple, fades to a brownish yellow, and usually disappears within 2–3 weeks. These colour changes may be more difficult to assess in people with darker skin tones.
    - The colour of a bruise is not a reliable indicator of the age of a bruise, and there is a wide variability in the appearance and healing of bruises, both among individuals and among bruises on the same person.
- Easy bruising implies that no significant trauma has occurred to the skin or subcutaneous tissue, and bruises are larger or more frequently seen than normal.

Causes:

- Bruising usually occurs as a result of trauma, which is usually accidental but may be non-accidental.
    - Bruising is the most common injury sustained by children who have been subject to physical abuse.
- Excessive bruising, or bruising that occurs as a result of minimal or no recognized trauma, may be caused by, or exacerbated by, an underlying bleeding disorder or medical condition, including:
    - Vascular disorders
    - Platelet disorders
    - Coagulation disorders
    - drugs
- Be aware that the presence of a bleeding disorder or other underlying medical condition does not rule out non-accidental injury as a cause of abnormal bruising, as the two may co-exist.

Trauma

- **Trauma-related bruising may be caused by:**
    - Accidental injury (which may or may not be recalled by the person or their carer). Bruises from accidental trauma in children tend to be:
        - Small (less than 15 mm in diameter).
        - Oval to round in shape with non-distinct borders.
        - Located above or near bony prominences on the front of the body (such as the forehead, knees, or shins).
    - Non-accidental injury — This may include:
        - Child maltreatment.
        - Domestic violence.
        - Self-inflicted injury (commonly on the legs or areas within easy reach).

- Complementary therapies — such as cupping (where cups create local suction on the skin, to promote healing) or coining (where coin rubbing causes dermabrasion, in the belief that this will relieve symptoms such as myalgia and abdominal pain).

Vascular disorders:

- **Vascular disorders leading to easy bruising include:**
  - **Senile purpura**
    - A common and benign condition, resulting from impaired collagen production and capillary fragility in elderly people.
    - Manifests as bruises that are usually bilateral on the extensor surfaces of the hands, forearms, face, and neck, which fade to a brownish colour over several months.
  - **Simple purpura ('easy bruising syndrome')**
    - A benign disorder typically occurring in otherwise healthy women, usually in their twenties or thirties.
    - Manifests as bruising on exposed areas (such as the arms or legs) after minor trauma.
    - May be associated with menorrhagia.
  - **Hereditary haemorrhagic telangiectasia (HHT)**
    - A rare autosomal dominant genetic disorder that leads to mucocutaneous telangiectasia of the skin, mucous membranes, and organs.
    - There may be a positive family history, recurrent nosebleeds, fatigue, nail changes, and hair loss.
  - **Ehlers-Danlos syndrome**
    - This causes joint hypermobility and skin translucency and hyperextensibility. It may present with gingival bleeding, prolonged bleeding after surgical procedures, and menorrhagia.
    - Abnormalities in capillary structure and deficiencies of perivascular collagen cause vascular fragility, and blood vessels may rupture when subject to shearing forces.
  - **Osteogenesis imperfecta**
    - This is characterized by blue sclerae, short stature, bone fragility, dentinogenesis imperfecta, and adult hearing loss.
    - Easy bruising after minimal or no trauma is thought to occur due to platelet dysfunction and capillary fragility.
  - **Vitamin C deficiency (scurvy)**
    - May occur following a lack of fresh fruit and vegetables over prolonged periods of time.
    - May present with perifollicular haemorrhage and bruising, dental deterioration, impaired wound healing, and coiled hairs.

Platelet disorders:

- **Platelet disorders leading to easy bruising include:**

- **Acute idiopathic thrombocytopenic purpura (ITP)**
- **Chronic ITP**
  - Most common in adult women.
  - Rarely undergoes spontaneous remission.
- **Henoch-Schlonlein purpura (HSP)**
- **Aplastic anaemia**
- **Malignancy with bone marrow involvement**
  - Acute lymphocytic or myelogenous leukaemia often present with fatigue, recurrent infections, and mucosal bleeding or bruising. Full blood count may show pancytopenia with thrombocytopenia, neutropenia, anaemia, and an elevated white blood cell count.
  - Neuroblastoma in a child, where tumour infiltration of the bone may present with periorbital bruising. The child is likely to be systemically unwell and may have a palpable abdominal mass.
  - Myeloma or myeloproliferative disorders may present with bone pain, fatigue, and anaemia.
- **End-stage chronic kidney disease**
  - May cause abnormalities of platelet aggregation and prolongation of bleeding time.
- **Liver disease**
  - May cause abnormalities of platelet function and number, the quality of clotting factors and proteins, and the quality of the skin and connective tissues.

Coagulation disorders:

- **Coagulation disorders are caused by a reduction or inhibition of circulating clotting factors, and may be inherited or acquired. Be aware that a negative family history does not exclude a genetically inherited disorder. Coagulation disorders leading to easy bruising include:**
  - **Haemophilia A (factor VIII deficiency) and haemophilia B (factor IX deficiency)**
  - **Impaired synthesis of clotting factor due to liver disease**
    - May be caused by drugs such as alcohol.
  - **Vitamin K deficiency**
  - **Von Willebrand disease (VWD)**
  - **Amyloidosis**
    - Abnormal bleeding is sometimes observed in people with immunoglobulin light chain (AL) amyloidosis.
    - Underlying mechanisms are thought to include prolonged prothrombin time, elevated plasmin-α2-antiplasmin complex, and factor X deficiency.

Drugs:

- **Corticosteroids (endogenous, topical, or oral)**
  - Corticosteroids can alter dermal blood vessels and cause collagen degradation and skin atrophy. Subsequent trauma or shearing stress results in purpura, which may be minor and appear to occur spontaneously.
  - Purpura typically occurs on exposed areas of the hands and forearms, or on the legs; are common in older people; are dark purple, and do not show the sequential colour changes of a normal bruise; and may persist for several weeks.
- **Drugs that inhibit platelet function**
  - Aspirin.
  - Nonsteroidal anti-inflammatory drugs.
  - Clopidogrel.
  - Selective serotonin reuptake inhibitors (SSRIs).
- **Drugs that can cause low platelets (thrombocytopenia)**
  - Alcohol.
  - Cephalosporins, nitrofurantoin, penicillins, sulfonamides.
  - Carbamazepine and valproic acid.
  - Quinine.
  - Propranolol.
  - Thiazide diuretics.
- **Drugs that inhibit coagulation inhibition**
  - Warfarin.
  - Heparins.
  - Apixaban, dabigatran, and rivaroxaban.

Diagnosis:

How should I assess a person with bruising?

- **Ask about symptoms that suggest an underlying platelet or coagulation disorder:**
  - Nosebleeds or gingival bleeding (mucocutaneous bleeding).
  - Excessive or prolonged bleeding from haemorrhoids, other rectal bleeding, haematuria, or menorrhagia.
  - Previous excessive bruising, or excessive or prolonged bleeding, that:
    - Occurs soon after trauma (particularly if it is associated with a petechial rash or mucocutaneous bleeding) — suggests a platelet disorders.
    - Is delayed, such as haemorrhage occurring 24 hours after a dental extraction (particularly if it is associated with bruises, haemarthrosis, or muscle haematomas) — suggests a coagulation disorder, such as haemophilia.
    - Is new in onset, following previously normal responses to trauma — suggests an acquired problem.

- **Ask about the person's general health to assess for underlying medical causes.**
    - A history of childhood chemotherapy or radiotherapy may result in a bone marrow disorder (such as myelodysplasia or leukaemia).
    - Hypothyroidism may affect the quality of skin and subcutaneous tissue.
    - Nutritional status:
        - Children who eat a limited diet can develop nutritional deficiencies, leading to coagulopathy, vascular fragility, and abnormal bruising.
        - Malnutrition can occur in older people, especially if living alone or in a nursing home, and in people with poorly fitting dentures.
    - Tiredness, weight loss, fever, and night sweats may suggest malignancy.
    - Joint pain, swelling, or reduced range of movement may suggest haemarthrosis.
- **Ask about alcohol use, and any prescribed or over-the-counter drugs.**
- **Ask whether there is a family history of:**
    - A known bleeding disorder (such as haemophilia, von Willebrand disease, or a platelet disorder).
    - A tendency to bruise or bleed easily or spontaneously.
    - Menorrhagia or postpartum bleeding in females, which may indicate a non-sex-linked disease (such as von Willebrand disease or factor XI deficiency).
    - Consanguinity — have a lower threshold for suspecting an autosomal recessively inherited bleeding disorder, such as factor X deficiency.
- **For infants, children, the elderly, or people with learning difficulties, assess bruising in the context of the person's age, mobility and developmental status, and the explanation for injury (if any). For children, ask if:**
    - The child is crawling — bruising is uncommon in infants who are not yet mobile.
    - The child may have taken a drug (such as warfarin) accidentally.
    - There is any history at birth of conditions suggesting an undiagnosed bleeding disorder, such as:
        - Cephalhaematoma after instrumental delivery.
        - Unexpected bleeding from the umbilical stump or delayed stump separation by up to 4 weeks, suggesting factor XIII deficiency.
        - Haematoma after routine intramuscular vitamin K given at birth.
        - Bleeding from the newborn heel prick test, suggesting factor XIII deficiency, or sometimes haemophilia.
- **Assess the location and pattern of bruising,** using pictorial or photographic records where possible and appropriate (for example in children):
    - Distribution, number, site, shape, and measured size of bruises.
    - Dependent areas — suggests thrombocytopenia or stasis factors.
    - Atypical areas, such as on the trunk — suggests an underlying bleeding disorder, or non-accidental injury.
    - Patterned bruising, for example, the outline of a hand print or implement (such as a belt).

- Only on the arms or legs — suggests trauma or changes in the skin or subcutaneous tissue.
- Periorbital — may suggest neuroblastoma in a child or amyloidosis (rare).
- Dorsum of the hands, extensor surface of the forearms, and the shins — suggests senile purpura.
- Note: the age of a bruise cannot be estimated accurately by its colour; if no bruises are currently present, ask the person to return when they reappear.

- **Examine the skin, hair, and nails for:**
  - Age-related changes.
  - Evidence of delayed healing, such as multiple scars or unhealed wounds — suggests corticosteroid use, hypothyroidism, ageing, self-inflicted injury, a collagen defect, or factor XIII deficiency.
  - Laxity — may suggest Ehlers danlos syndrome
  - Pallor — suggests anaemia, which may be associated with malignancy.
  - Jaundice — suggests liver disease.
  - Petechiae (tiny, round, non-blanching, pinpoint flat spots less than 3 mm in diameter) — for example at clothing line pressure sites may indicate a platelet disorder; or if in the distribution of the superior vena cava they may follow coughing, vomiting, or strangulation.
  - Palpable purpura (typically 3–10 mm in diameter) — suggests an underlying systemic vasculitis, such as Henoch schonlein purpura
  - Brittle hair and nails — suggests nutritional factors, ageing, hypothyroidism, or rarely hereditary haemorrhagic telangiectasia.
- **Examine the joints for:**
  - Hyperextensibility or elasticity — suggestive of Ehlers danlos syndrome
  - Swelling and tenderness — may indicate haemarthrosis, suggesting hemophilia
  - Tenderness — may be seen in acute leukaemia or neuroblastoma.
- **Examine the abdomen for:**
  - Splenomegaly — suggests malignancy or ITP (rare).
  - Hepatomegaly — suggests malignancy or liver disease.
  - Ascites, caput medusa, and spider telangiectasia — suggests chronic liver disease.
- **Examine the head and neck.**
  - The oropharynx — for signs of bleeding, trauma, or healing injury to the frenulae (may suggest other non-accidental injury); gum hypertrophy may occur in monocytic leukaemia; wet purpura on the buccal mucosa or tongue is suggestive of severe thrombocytopenia.
  - The eyes — use fundoscopy to check for retinal haemorrhages (may suggest other non-accidental injury).
- **Examine for lymphadenopathy.**
  - Suggestive of leukaemia or amyloidosis

Red flags suggesting non-accidental injury:

- **Suspect non-accidental injury and physical abuse in infants, children, the elderly, people with learning difficulties, and otherwise vulnerable people when:**
  - **Bruises are on a child who is not yet independently mobile (crawling, cruising, or walking).**
    - Bruising appropriate to learning to walk is common from around 1 year of age when most children have started 'cruising'. It is typically distributed on the anterior tibia and knee, followed by the forehead.
  - **Bruises have indicative features.**
    - Disproportionate to the explanation of injury sustained.
    - Unusually large.
    - Present in multiple sites or in clusters.
    - Of a similar shape and size.
    - Patterned in the shape of a hand print, ligature, stick, tooth (or teeth marks), grip, or implement (such as a belt). Fingertip bruising is often found in children with a bleeding disorder.
    - Associated with petechiae.
  - **Bruises are found in indicative places.**
    - Sites that are not typical for the age of a child.
    - Any non-bony part of the body or face (including the eyes, ears, cheeks, back, abdomen, buttocks, arms, and genitalia).
    - On both sides of the face or head.
    - On the neck (consistent with strangulation).
    - On the ankles and wrists (consistent with use of a ligature).
  - **The explanation for the bruising is implausible, inadequate, or inconsistent:**
    - With a person's presentation, normal activities, existing medical condition, age or developmental stage, or account — compared with that given by parents or carers.
    - Between parents or carers.
    - Between accounts over time.
  - **There is a delay in presentation.**

How should I investigate a person with bruising?

- **Arrange a urine dipstick test** to check for non-visible haematuria that may suggest an underlying bleeding disorder or vasculitis.
- **In adults with unexplained bruising, bleeding, petechiae, or hepatosplenomegaly consider arranging:**
  - A very urgent full blood count including platelet count (within 48 hours) and blood film.
  - Clotting screen (prothrombin time [PT], activated partial thromboplastin time [aPTT], and international normalized ratio [INR] if the person is taking warfarin).
  - Liver, renal, and thyroid function tests, depending on clinical judgement.

- **In children and young people with unexplained bruising or bleeding offer:**
    - A very urgent full blood count including platelet count (within 48 hours) and blood film.
- Be aware that blood results such as full blood count and clotting screen can be normal in people with mild coagulation and platelet function disorders.

Differential diagnosis:

Skin conditions that might be mistaken for bruising include:

- **Linear eruptions**
    - Allergic contact dermatitis.
    - Phytophotodermatitis — an inflammatory and pigmentary reaction of the skin to light, following contact with phototoxic substances from certain plants, such as giant hogweed.
    - Stretch marks (striae).
- **Non-linear eruptions**
    - Slate-grey naevi (Mongolian blue spots) — macular blue-grey non-tender macules present at birth on the sacrum and buttocks in normal dark-skinned infants.
    - Haemangiomas — benign congenital cutaneous tumours.
    - Urticaria pigmentosa — numerous reddish-brown or pale maculopapules, plaques, or nodules that urticate within a few minutes of gentle rubbing. They are distributed symmetrically usually on the trunk and thighs, but can present anywhere on the body, sparing the face, palms, and soles.
    - Dermatitis artefacta — self-inflicted skin lesions.
    - Congenital vascular naevi.
- **Other**
    - Skin staining from dye or ink.

Management:

From age 1 month onwards.
When should I admit or refer a person with abnormal bruising?

- **Admit the person if they have significant active bleeding** and are not responding to simple measures (such as local compression).
- **If non-accidental injury is suspected in children and young people, the elderly, or other vulnerable people:**
    - Refer immediately to the most appropriate agency
- **If non-accidental injury is not suspected:**
    - **Refer children and young people for immediate specialist assessment** for leukaemia if they have:
        - Unexplained petechiae, or
        - Hepatosplenomegaly, or

        - Full blood count results suggestive of leukemia.
    - **Refer children with periorbital bruising urgently (for an appointment within 48 hours) for specialist assessment** for neuroblastoma if they have:
        - A palpable abdominal mass, or
        - An unexplained enlarged abdominal organ.
    - **For all other children,** liaise with a paediatrician or paediatric haematologist to arrange venepuncture and further investigations as needed.
    - **Refer adults for immediate specialist assessment** for leukaemia if they have:
        - Full blood count results suggestive of leukemia.
    - **Refer all other adults to a haematologist** for further investigations, the urgency depending on clinical judgement, if there is:
        - A low platelet-count.
        - An abnormal clotting screen.
        - Normal blood results in primary care but a bleeding disorder is still suspected.

How should I manage bruising in primary care?

- **For simple bruising where there is no suspected underlying bleeding disorder:**
    - Prescribe simple analgesia such as paracetamol if required.
- **For people taking warfarin with an abnormal clotting screen (a prolonged prothrombin time or increased international normalized ratio):**
    - If the person is monitored and managed in primary care, alter their warfarin dosage according to local protocols, or liaise with the local warfarin clinic or a haematologist, depending on clinical judgement.
- **For people taking a drug that may cause thrombocytopenia,** withdraw the drug where possible and appropriate, and monitor the person for resolution of symptoms and signs.

TOPIC: Antiphospholipid syndrome

Clinical features:

In addition to the hallmark features of venous, arterial, and/or small vessel thrombosis as well as specific pregnancy complications, other relatively common clinical features of antiphospholipid syndrome (APS) include livedo reticularis, thrombocytopenia, or transient ischemic attack.

Thrombotic events — Thromboses are the hallmark of APS, and venous thromboses are more common than arterial thromboses

- **Venous thrombosis** — The deep veins of the lower extremities are the most common sites of thrombosis.
- **Arterial thrombosis** — The most common site of arterial thrombosis is in the cerebral vasculature, usually in the form of a stroke or transient ischemic attack. Occlusions in the retinal, coronary, renal, and mesenteric arteries can also occur

Neurologic involvement — Central nervous system abnormalities are a common feature of APS that have been attributed to both vascular thrombosis as well as direct injury to neuronal tissue by aPL. Stroke and transient ischemic attack are the most common neurologic manifestations of APS

Hematologic abnormalities — Thrombocytopenia can be observed in APS

Pulmonary involvement — Patients with APS may develop various lung manifestations including pulmonary thromboembolic disease, thromboembolic and non-thromboembolic pulmonary hypertension (pulmonary arterial hypertension, pulmonary arterial thrombosis.

Cardiac involvement — Cardiac manifestations of APS most commonly involve the valves, including valvular thickening and valve nodules

Libman-Sacks endocarditis and cardiac valve disease:

- Usually, mitral valve disease or aortic valve disease and usually regurgitation with or without stenosis

Cutaneous manifestations — APS has been associated with many cutaneous abnormalities including splinter hemorrhages, livedo reticularis

History:

The history should focus on the nature and frequency of thrombotic events, the outcomes of pregnancies in females, thrombocytopenia, and other risk factors for thrombosis, which may include immobility, use of estrogen-containing medications, and/or family history of thrombophilia. A history of heparin exposure may be relevant in patients with possible heparin-induced thrombocytopenia (HIT). The history should also include inquiries about symptoms associated with systemic lupus erythematosus (SLE) such as photosensitivity, oral ulcers, patchy hair loss, and Raynaud phenomenon.

Investigations:

Antiphospholipid antibodies — The three major aPL tests that are recognized by international classification criteria for antiphospholipid syndrome (APS) are as follows:

- Anticardiolipin antibodies (aCL) immunoglobulin G (IgG) and/or IgM enzyme-linked immunosorbent assay (ELISA)
- Anti-beta2 glycoprotein (GP) I antibodies IgG and/or IgM ELISA
- Lupus anticoagulant (LA) test

Management: based on EULAR guidelines:

Advice:

A healthy lifestyle in line with prevention of cardiovascular disease is recommended:

- Avoid smoking.
- Take regular physical exercise.
- Maintain a healthy diet and avoid overweight/obesity.
- Avoid excessive alcohol intake.
- Adequate management of cardiovascular risk factors, including diabetes, hypertension and hyperlipidaemia

prescribe:

- primary thromboprophylaxis
  - low-dose aspirin
- secondary thromboprophylaxis
  - initial venous thromboembolic events: lifelong warfarin with a target INR of 2-3
  - recurrent venous thromboembolic events: lifelong warfarin; if occurred whilst taking warfarin then consider adding low-dose aspirin, increase target INR to 3-4
  - arterial thrombosis should be treated with lifelong warfarin with target INR 2-3

TOPIC: DVT/DVT prevention for travelers

Definition:

- Deep vein thrombosis (DVT) is the term used to describe the formation of a thrombus (blood clot) in a deep vein, which partially or completely obstructs blood flow.
- DVT can occur as a result of prolonged immobility during long distance travel (for example, by bus, train, or air).
    - Prolonged cramped sitting during long-distance travel interferes with venous flow in the legs and causes venous stasis.
    - Seat-edge pressure on the popliteal area may contribute to vessel wall damage as well as venous stasis.
    - Coagulation activation may result from an interaction between cabin conditions (such as hypobaric hypoxia) and individual risk factors for venous thromboembolism (VTE).

What are the risks of developing travel-related deep vein thrombosis?

- The risk of venous thromboembolism (VTE) decreases with time after air travel and returns to baseline by eight weeks.
    - Most air travel-related VTE occurs within the first 1–2 weeks after the flight.
- The risk of travel-related DVT increases in the presence of other risk factors.

What are the risk factors for travel-related deep vein thrombosis?

The relative risk of travel-related deep vein thrombosis (DVT) is further increased if other risk factors for developing DVT are present. These include:

- Active malignancy.
- A family history of DVT or pulmonary embolism (PE) in a close family member.
- Aged over 60 years.
- Extremes of height — people taller than 1.90 metres or shorter than 1.60 metres.
    - Shorter people may experience seat edge pressure to the popliteal area.
    - Taller people have less leg room.
- Inherited blood-clotting abnormalities.
- Large varicose veins or chronic venous insufficiency.
- Limited mobility (for example, a lower-limb fracture in plaster).
- Obesity (body mass index greater than 30 kg/m2).
- Polycythaemia.
- Pregnancy, or up to six weeks postpartum.
- Previous DVT or PE.
- Recent surgery or trauma, particularly to the abdomen, pelvic region, or legs.

- Thrombophilia (such as factor V Leiden mutation or antiphospholipid syndrome), or other thrombotic states (for example, heart failure, recent myocardial infarction, severe infection, inflammatory bowel disease).
- Use of oestrogens, such as oral contraceptives or hormone replacement therapy.

Management:

Deep vein thrombosis - prevention for travelers

From age 16 years onwards.
How should I assess DVT risk in travelers?

- Assessment should be made on an individual basis as the relative risk of deep vein thrombosis (DVT) may change over time.
- Take into account the planned duration of travel and any pre-existing risk factors for DVT.
- **People are at:**
  - **High risk of travel-related DVT if they have:**
    - An active malignancy.
    - Undergone recent major surgery (within the previous 4 weeks).
    - Had a previous unprovoked venous thromboembolism (VTE).
    - Had a previous travel-related DVT with no associated temporary risk factor.
    - More than one risk factor for DVT.
  - **Moderate risk** if they have a risk factor and are not considered at high risk.
  - **Low risk** if they have no pre-existing risk factors.

What advice should I give someone who is at low risk of travel-related DVT?

- **For people assessed as having a relatively** low risk **of developing travel-related deep vein thrombosis (DVT):**
  - Provide advice on general measures to reduce the risk of travel-related DVT.
  - Reassure the person that no specific treatment is required.

What advice should I give someone who is at moderate risk of travel-related DVT?

- **For people at** moderate risk **of developing travel-related deep vein thrombosis (DVT):**
  - Provide advice on general measures to reduce the risk of travel-related DVT.
  - Advise the person to use graduated compression stockings providing 15–30 mm Hg of pressure at the ankle.
    - Class 1 stockings (exerting a pressure of 14–17 mmHg at the ankle) or proprietary flight socks may be sufficient.
    - Measure the ankle-brachial pressure index (ABPI) in both legs using a Doppler machine if the person has symptoms of arterial disease.

What advice should I give someone who is at high risk of travel-related DVT?

- **For people assessed at high risk of developing travel-related deep vein thrombosis (DVT):**

    - Assess the person's suitability for long distance travel. If appropriate:
        - Consider seeking specialist advice, or
        - Recommend delaying or cancelling the trip (for example, people should postpone long-haul flights for 3 months after a hip or knee replacement).
- **If long distance travel is unavoidable:**
    - Provide advice on general measures to reduce the risk of travel-related DVT.
    - Advise the use of graduated compression stockings providing 15-30 mm Hg of pressure at the ankle.
    - Seek specialist advice depending on local pathways regarding whether the use of low molecular weight heparin is indicated.

What general advice should I give to all long-distance travelers?

- Offer reassurance that the risk of travel-related deep vein thrombosis (DVT) is very low in healthy people without risk factors.
- **Advise the person to:**
    - Avoid long periods of immobility. Walk around regularly — for example, a regular trip to the bathroom every 2–3 hours.
    - Choose an aisle seat when feasible.
    - Perform frequent calf muscle exercises.
    - Ensure hand luggage is not placed somewhere that it restricts movement of legs and feet.
    - Wear clothing that is comfortable and loose.
    - Avoid the use of tranquilisers, or sleeping tablets.
    - Maintain a normal fluid intake (at least 250 ml every 2 hours) and avoid excessive alcohol consumption.
- Seek urgent medical advice (for example, from a local doctor or the nearest Accident and Emergency department) if they develop the following after the trip:
    - Swollen, painful legs, especially where one is more affected than the other, and/or
    - Breathing difficulties (suggesting pulmonary embolism).
- Obtain adequate medical insurance before they travel.

How should I manage a person with a joint replacement or fracture?

- **For people who have had a hip or knee replacement:**
    - Advise them to postpone long-haul flights until 3 months after surgery as there may be an increased risk of deep vein thrombosis.
    - Advise them that it may be possible to undertake short-haul flights 6 weeks after surgery.
- **For people with a fracture in a plaster cast:**
    - Advise people that airlines usually restrict air travel on short flights (less than 2 hours) for the first 24 hours after a cast has been fitted, and for 48 hours for longer flights (over 2 hours).

    - If air travel is essential before these limits, airlines usually require the cast to be split along its full length before travel.
        - Arrangements should be made to have the cast replaced once at the destination.

How should I manage a person with a recent deep vein thrombosis planning long-haul travel?

- **If the person has a new diagnosis (within 2 weeks) of deep vein thrombosis (DVT) or pulmonary embolism, seek specialist advice.**
- **If the person has been taking anticoagulants for 2 weeks or more:**
    - Reassure them that they are at low-risk of developing a further thrombosis.
    - Provide advice on general measures to reduce the risk of travel-related DVT.
    - Consider prescribing graduated compression stocking.

What advice should I give about graduated compression stockings?

- Graduated compression stockings are not available on prescription for the sole indication of preventing travel-related DVT. However, they can be prescribed for people with the following indications:
    - Class 1 stockings (compression at the ankle 14–17 mmHg) for:
        - Superficial or early varicose veins.
        - Varicose veins during pregnancy.
    - Class 2 stockings (compression at the ankle 18–24 mmHg) for:
        - Varicose veins of medium severity.
        - Treatment and prevention of recurrence of leg ulcers.
        - Mild oedema.
        - Varicose veins during pregnancy.
    - Class 3 stockings (compression at the ankle 25–35 mmHg) for:
        - Gross varicose veins.
        - Post-thrombotic venous insufficiency.
        - Gross oedema.
        - Treatment and prevention of recurrence of leg ulcers.

Should I recommend aspirin for the prevention of travel-related deep vein thrombosis?

- Aspirin is not recommended for the prevention of travel-related deep vein thrombosis.
- People already taking aspirin (for primary or secondary prevention of cardiovascular disease) should not increase their dose.
    - If low molecular weight heparin is prescribed, aspirin should not be discontinued without specialist advice.

TOPIC: Post-thrombotic syndrome

It is increasingly recognised that patients may develop complications following a DVT. Venous outflow obstruction and venous insufficiency result in chronic venous hypertension. The resulting clinical syndrome is known as post-thrombotic syndrome. The following features maybe seen:

- painful, heavy calves
- pruritus
- swelling
- varicose veins
- venous ulceration

Compression stockings have in the past been offered to patients with deep vein thrombosis to help reduce the risk of post-thrombotic syndrome.

However, Clinical Knowledge Summaries now state the following:

Do not offer elastic graduated compression stockings to prevent post-thrombotic syndrome or VTE recurrence after a proximal DVT. This recommendation does not cover the use of elastic stockings for the management of leg symptoms after DVT.

However, once post-thrombotic syndrome has developed compression stockings are a recommended treatment. Other recommendations including keeping the leg elevated.

TOPIC: Haematological cancers - recognition and referral

Presentation:

- **Leukaemia:**
  - Common signs of leukaemia include anaemia, reduced normal white cells, and thrombocytopaenia.
  - Symptoms of leukaemia include pallor, bruising, and a propensity to infection.
  - Many chronic leukaemias are symptomless and are only identified when a full blood count is performed for other reasons.
- **Myeloma:**
  - Symptoms include bone pain, often in multiple sites such as the ribs, and bone marrow failure.
  - Complications include kidney failure and thrombo-embolism.
- **Non-Hodgkin's lymphoma:**
- **Hodgkin's lymphoma:**

Prognosis:

- Leukaemia: most forms of leukaemia have high 5 years survival, though some subtypes have a poorer prognosis.
- Myeloma: 5 years survival is nearly 50%.
- Non-Hodgkin's lymphoma: 5 years survival is just under 70%.
- Hodgkin's lymphoma: 5 years survival is 85%.

Management:
Referral for hematological cancer:
When should I refer an adult with suspected leukaemia?

- **Consider a very urgent full blood count** (within 48 hours) to assess for leukaemia in adults with any of the following:
  - Pallor.
  - Persistent fatigue.
  - Unexplained fever.
  - Unexplained persistent or recurrent infection.
  - Generalized lymphadenopathy.
  - Unexplained bruising.
  - Unexplained bleeding.
  - Unexplained petechiae.
  - Hepatosplenomegaly.

When should I refer an adult with suspected myeloma?

- **Offer a full blood count, blood tests for calcium and plasma viscosity or erythrocyte sedimentation rate** to assess for myeloma in people aged 60 years and over with persistent bone pain, particularly back pain, or unexplained fracture.

- **Offer very urgent protein electrophoresis and a Bence-Jones protein urine test** (within 48 hours) to assess for myeloma in people aged 60 years and over with hypercalcaemia or leukopenia and a presentation that is consistent with possible myeloma.
- **Consider very urgent protein electrophoresis and a Bence-Jones protein urine test** (within 48 hours) to assess for myeloma if the plasma viscosity or erythrocyte sedimentation rate and presentation are consistent with possible myeloma.
- **Refer people using a suspected cancer pathway referral** (for an appointment within 2 weeks) if the results of protein electrophoresis or a Bence-Jones protein urine test suggest myeloma.

When should I refer an adult with suspected non-Hodgkin's lymphoma?

- **Consider a suspected cancer pathway referral** (for an appointment within 2 weeks) for non-Hodgkin's lymphoma in adults presenting with unexplained lymphadenopathy or splenomegaly. When considering referral, take into account any associated symptoms, particularly fever, night sweats, shortness of breath, pruritus, or weight loss.

When should I refer a child or young person with suspected non-Hodgkin's lymphoma?

- **Consider a very urgent referral** (for an appointment within 48 hours) for specialist assessment for non-Hodgkin's lymphoma in children and young people presenting with unexplained lymphadenopathy or splenomegaly. When considering referral, take into account any associated symptoms, particularly fever, night sweats, shortness of breath, pruritus, or weight loss (new NICE recommendation for 2015).

When should I refer an adult with suspected Hodgkin's lymphoma?

- **Consider a suspected cancer pathway referral** (for an appointment within 2 weeks) for Hodgkin's lymphoma in adults presenting with unexplained lymphadenopathy. When considering referral, take into account any associated symptoms, particularly fever, night sweats, shortness of breath, pruritus, weight loss, or alcohol-induced lymph node pain.

When should I refer a child or young person with suspected Hodgkin's lymphoma?

- Consider a very urgent referral (for an appointment within 48 hours) for specialist assessment for Hodgkin's lymphoma in children and young people presenting with unexplained lymphadenopathy. When considering referral, take into account any associated symptoms, particularly fever, night sweats, shortness of breath, pruritus, or weight loss.

Safety netting

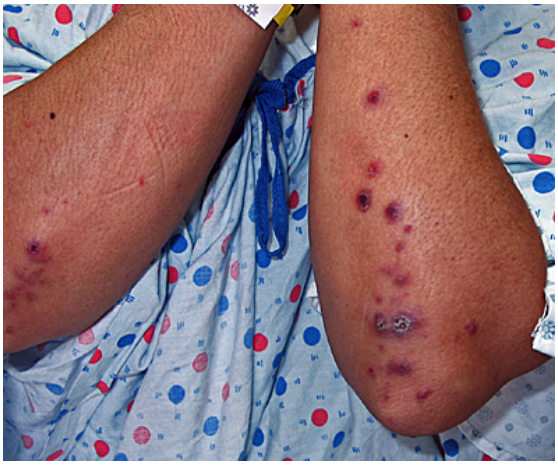
- Safety netting involves active monitoring of low-risk people presenting with symptoms in primary care.
- Explain to people who are being offered safety netting which symptoms to look out for and when they should return for re-evaluation. It may be appropriate to provide written information.
- Reassure people in the safety netting group who are concerned that they may have cancer that with their current symptoms their risk of having cancer is low.
- Ensure that the results of investigations are reviewed and acted upon appropriately, with the healthcare professional who ordered the investigation taking or explicitly passing on responsibility for this. Be aware of the possibility of false-negative results for chest X-rays and tests for occult blood in faeces.

TOPIC: Acute myeloid leukemia

AML is the second most common category of leukemia in adults and the most common type of acute leukemia. Nevertheless, AML is relatively rare and it accounts for approximately 1 percent of adult cancers in the United States, but nearly 2 percent of cancer-related deaths. The median age at diagnosis is approximately 68 years, and the incidence increases with age.

CLINICAL PRESENTATION

- Patients with AML generally present with symptoms related to complications of pancytopenia (eg, anemia, neutropenia, and thrombocytopenia).
- Neutropenia: fever and recurrent infection
- Signs and symptoms related to anemia (eg, shortness of breath, weakness, dyspnea), thrombocytopenia (excess bleeding or bruising), and neutropenia (fever, infections) are common.
- Thrompocytopenia: hemorrhagic findings such as gingival bleeding, ecchymoses, epistaxis, or menorrhagia
- Fatigue is present in the majority of patients and often precedes the diagnosis for a number of months.
- **Skin** — Examination of the skin can reveal pallor secondary to anemia, petechiae or ecchymoses secondary to thrombocytopenia or disseminated intravascular coagulation, or infiltrative lesions suggestive of leukemic involvement (leukemia cutis or myeloid sarcoma)).



Oropharynx — Careful examination of the oropharynx and teeth may reveal leukemic involvement (eg, gingival hypertrophy, especially in the monocytic subtypes)

Organomegaly — Palpable adenopathy is uncommon in patients with AML and significant lymph node enlargement is rare. Similarly, hepatomegaly and splenomegaly are present in approximately 10 percent.

- Bone marrow aspiration and biopsy (usually unilateral) is a key component to the diagnosis of AML

Diagnosis

Bone marrow aspiration and biopsy is the diagnostic procedure

The diagnosis of AML requires both of the following:

- Documentation of bone marrow infiltration – Blast forms must account for at least 20 percent of the total cells of the bone marrow aspirate
- The leukemic cells must be of myeloid origin as demonstrated by either the presence of Auer rods

Evaluation

•**do urgent Complete blood count (CBC)** – CBC usually reveals leukocytosis due to malignant blasts, but the leukocyte count may be low or normal with few or no blasts are

detected; most patients have a normochromic, normocytic anemia, neutropenia and thrombocytopenia.

•**Blood smear**

•**Bone marrow (BM) examination** – BM is generally hypercellular with ≥20 percent blasts and replacement of normal, maturing cells. There may be substantial fibrosis in some cases.

●**Diagnosis** – AML is generally diagnosed by morphologic, cytochemical, immunophenotypic, and cytogenetic/molecular analysis of bone marrow; in some cases, AML can be diagnosed from blood or a chloroma.

Differential diagnosis:
aplastic anemia, nutritional deficiencies, myelofibrosis lymphoid leukemia, chronic myeloid leukemia, myelodysplastic syndromes (MDS)

Management:

If you suspect leukemia do urgent cbc within 48 hours.

If Cbc suspicious of leukemia refer to hematology within 2 weeks

Explain:

Acute myeloid leukemia (called "AML") is a type of blood cancer. AML is usually fast-growing and needs to be treated quickly.

Blood is made up of different types of cells. These cells are made in the center of your bones, in a part called the bone marrow. When people have AML, their bone marrow makes abnormal blood cells instead of normal blood cells

Treatment for AML usually includes 2 parts.

The first part of treatment is called "induction of remission" and lasts about 4 weeks. During this part, people stay in the hospital and get **chemotherapy**. Chemotherapy is the medical term for medicines that kill cancer cells or stop them from growing.

TOPIC: Chronic lymphocytic leukemia

CLL is the most common leukemia in Western countries. It has a male predominance and is more common in White Americans. The median age at diagnosis is approximately 70 years.

Clinical features:

Lymphadenopathy — The most common abnormal finding on physical examination of the patient with CLL is lymphadenopathy, present in 50 to 90 percent of patients among various series. Lymph node enlargement may be generalized or localized, and individual lymph nodes can vary greatly in size. Cervical, supraclavicular, and axillary lymph nodes are affected most commonly.

Splenomegaly — The spleen is the second most frequently enlarged lymphoid organ, being palpably enlarged in 25 to 55 percent of cases

Hepatomegaly — Enlargement of the liver may be noted at the time of initial diagnosis in 15 to 25 percent of cases

Laboratory abnormalities:

Lymphocytosis — The most noteworthy laboratory abnormality found in CLL is lymphocytosis in the peripheral blood and bone marrow.

Cytopenias — Neutropenia, anemia, and thrombocytopenia may be observed at the time of initial diagnosis and are usually not severe.

Peripheral smear — The peripheral blood smear of patients with CLL demonstrates lymphocytosis.

The smear also often contains "smudge" cells (also called "basket cells")

Immunophenotype — Immunophenotypic analysis, usually by flow cytometry, is a key component to the diagnosis of CLL

Bone marrow aspirate and biopsy — Bone marrow aspirate and biopsy are not required for the diagnosis of CLL.

Differential Diagnosis:

diagnosis of CLL is suspected whenever the peripheral blood in an adult demonstrates an absolute lymphocytosis. However, blood lymphocytosis may also occur with non-neoplastic conditions, such as viral or other infections (eg, infectious mononucleosis, pertussis, toxoplasmosis), as well as in neoplastic conditions other than CLL (eg, the leukemic phase of lymphomas, hairy cell leukemia, prolymphocytic leukemia, and large granular cell lymphocyte leukemia).

TOPIC: Burkitt's lymphoma

- Burkitt's lymphoma is a high-grade B-cell non-Hodgkin's lymphoma that is rapidly growing and aggressive.

- Burkitt's lymphoma is significantly more common in sub-Saharan Africa, where it accounts for approximately one half of childhood cancers. The incidence peaks at around 5-8 years of age and is more common in boys.

There are three main types of Burkitt's lymphoma:

Three distinct clinical forms of BL are recognized: endemic (African), sporadic (non-endemic), and immunodeficiency-associated

Clinical features

Endemic Burkitt's lymphoma

- Most commonly involves the maxilla and/or mandible; however, abdominal organs may also be involved.

- A tumour can double in size in 18 hours.
- Presentation is commonly with:
  - Swelling of the affected jaw/bones.

  - Rapidly enlarging, non-tender lymph nodes in the neck/jaw.

Sporadic Burkitt's lymphoma

- Usually involves the abdominal organs, most frequently the distal ileum, caecum or mesentery.

- Other abdominal and pelvic organs, glandular tissue (such as the thyroid and tonsils) and the facial bones may also be involved.
- Presentation is commonly with:
  - Abdominal mass/swelling and ascites.

  - Abdominal pain.

  - Intestinal obstruction.

Other presentations

- B symptoms may be present with both forms of Burkitt's lymphoma and include:

  - Fever

  - Weight loss

  - Night sweats
- HIV-related Burkitt's lymphoma tends to present in a similar way to the sporadic disease.

Diagnosis:

Biopsy is needed for diagnosis

- The diagnosis of BL is based on the pathologic evaluation of involved tissue, usually an abdominal mass.

Microscopy findings:

- 'starry sky' appearance: lymphocyte sheets interspersed with macrophages containing dead apoptotic tumour cells

- CXR, CT and MRI scanning allow staging.

TOPIC: Hodgkin lymphoma

- There is a peak in incidence in young adults aged 20-34 years, with a further peak observed over 70 years.

CLINICAL PRESENTATION:

Typical presentations — Most patients with classical HL(chL) present with asymptomatic lymphadenopathy or a mass on chest radiograph . Constitutional symptoms ("B" symptoms; ie, fever, night sweats, or unintended weight loss) are present in approximately 40 percent of cases.

Lymphadenopathy — Lymphadenopathy is detectable in more than two-thirds of patients with cHL at presentation, and the involved lymph nodes are usually nontender and have a firm, rubbery consistency.
The neck is the most common site of involvement, as 60 to 80 percent of patients have enlarged cervical and/or supraclavicular nodes. Enlarged axillary nodes are found in approximately 30 percent and inguinal nodes in 10 percent of patients

Mediastinal mass — Discovery of a mediastinal mass on routine chest radiograph is another common presentation of cHL.

the mass may be asymptomatic or associated with cough, shortness of breath, or retrosternal chest pain.

Other symptoms (eg, fatigue, pruritus, alcohol-associated pain) are not considered B symptoms.

Examples of less common clinical presentations of cHL include:

Liver disease – Liver involvement may be manifest as abnormal liver function tests or as abdominal pain, nausea, anorexia, other nonspecific findings

Bone/bone marrow involvement

Examples of laboratory findings at presentation include:

•**Hypercalcemia** – Hypercalcemia is usually due to increased production of calcitriol (1,25-dihydroxyvitamin D3) and less commonly is caused by direct bony involvement.
•**Anemia** – Anemia can be due to diverse causes, including bone marrow replacement by cHL, hypersplenism, anemia of chronic inflammation, and rarely may be due to a Coombs-positive hemolytic anemia, with or without immune thrombocytopenia
•**Eosinophilia** – Eosinophilia is relatively common in HL and is caused by production of chemokines (eg, interleukin-5, eotaxin) that recruit eosinophils and/or stimulate eosinophil production
•**Other** – Other laboratory abnormalities may include leukocytosis, thrombocytosis, lymphopenia, and hypoalbuminemia. Some of these findings are associated with inferior prognosis, as discussed separately.

- HIV tests are necessary in patients with suspected Hodgkin's lymphoma.
- ESR: an ESR of greater than 70 carries an unfavourable prognosis.
- Liver function and serum protein tests: the level of any rise in lactate dehydrogenase (LDH) and fall in albumin levels has prognostic significance.

EVALUATION

Imaging — Imaging may be used to identify potential sites for biopsy and to evaluate organ involvement.

- **Biopsy** – An excisional biopsy (or multiple core needle biopsies) is required to diagnose cHL and determine the histologic subtype; a fine needle aspiration (FNA) is generally **not** sufficient for diagnosis and classification of cHL.

- **Lymphadenopathy** – For the patient who does not have accessible peripheral lymphadenopathy, ultrasonography, computed tomography (CT), or positron emission tomography (PET) may identify a suspicious site and/or guide a tissue biopsy.

Microscopy — In cHL, the lymph node is effaced by variable numbers of Hodgkin/Reed-Sternberg (HRS) cells admixed with a polymorphous inflammatory infiltrate.
Consider suspected cancer pathway referral (within 2 weeks) for adult with Hodgkin lymphoma in adults present with unexplained lymphadenopathy.

Treatment can include:

●Chemotherapy – Chemotherapy is the medical term for medicines that kill cancer cells or stop them from growing. Doctors can use different chemotherapy medicines to treat Hodgkin lymphoma. Your doctor will work with you to choose the ones that are right for you.
●Radiation therapy – Radiation kills cancer cells. It is used along with chemotherapy for some people with Hodgkin lymphoma

TOPIC: Non-Hodgkin's lymphoma

- Non-Hodgkin's lymphoma is more than five times as common as Hodgkin's disease.
- White people have a higher risk of developing non-Hodgkin's lymphoma than black and Asian people.
- The median age at presentation is older than 50 years, except for patients with high-grade lymphoblastic and small noncleaved lymphomas, which are the most common types of NHL observed in children and young adults. Low-grade lymphomas are very rare in children.

CLINICAL FEATURES

- **Low-grade lymphomas:**
  - Painless, slowly progressive peripheral lymphadenopathy is the most common clinical presentation. Spontaneous regression of enlarged lymph nodes may occur.
  - Primary extra-nodal involvement and systemic symptoms (fatigue, weakness, fever, night sweats, weight loss) are not common at presentation but are common in patients with advanced or end-stage disease.
  - Bone marrow is frequently involved and may be associated with cytopenia.
  - Splenomegaly.
  - Hepatomegaly.
- **Intermediate- and high-grade lymphomas:**
  - Most patients present with rapidly growing and bulky lymphadenopathy.
  - Systemic symptoms and extranodal involvement (especially the gastrointestinal tract, skin, bone marrow, sinuses, genitourinary tract, thyroid, and central nervous system (CNS)) is more common.
  - Hepatomegaly, splenomegaly.
  - Obstructive hydronephrosis secondary to bulky retroperitoneal lymphadenopathy obstructing the ureters may occur.
  - Primary CNS lymphomas: account for 1% of all intracranial neoplasms and are more commonly observed in patients with immunodeficiency - eg, Wiskott-Aldrich syndrome, transplantation or HIV infection.
  - Skin lesions: associated with cutaneous T-cell lymphoma (mycosis fungoides), anaplastic large-cell lymphoma and angio-immunoblastic lymphoma.
  - Lymphoblastic lymphoma: high-grade lymphoma, which often manifests with a mediastinal mass, superior vena cava syndrome and meningeal disease with cranial nerve palsies.
  - Burkitt's lymphoma: often presents with a large abdominal mass and symptoms of bowel obstruction

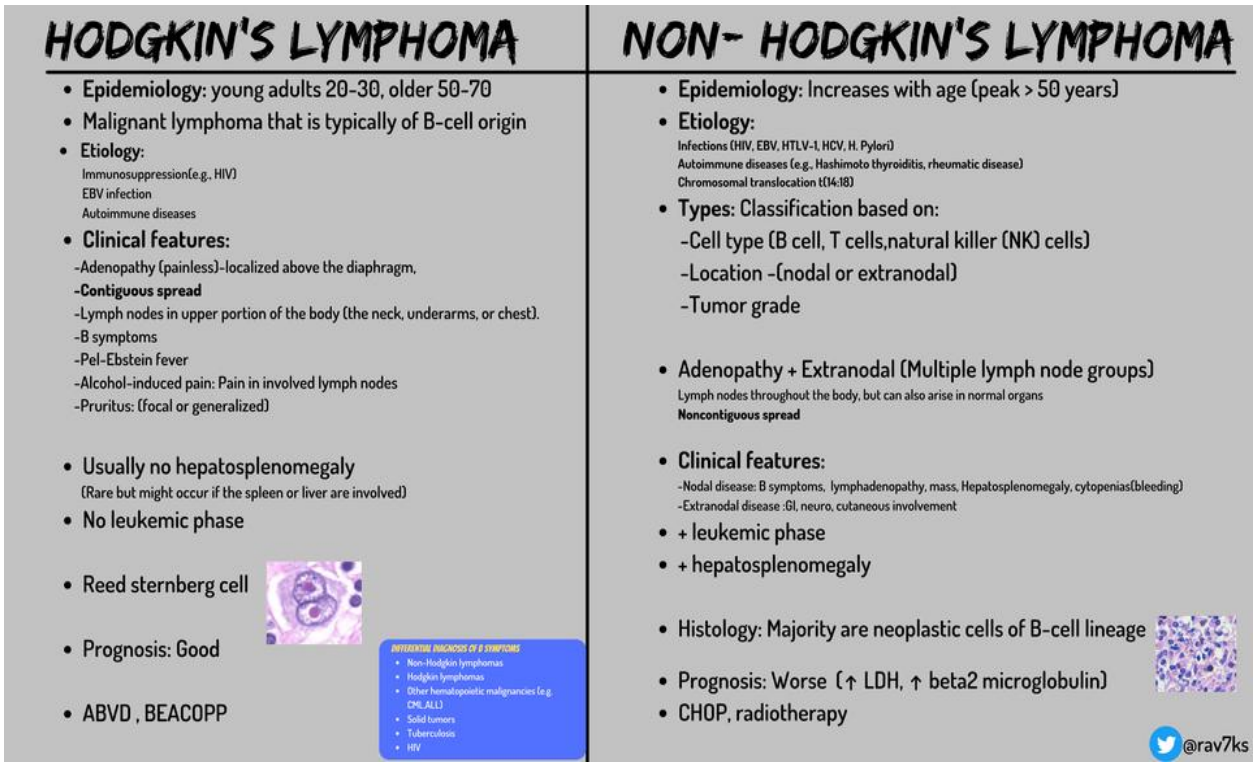
Risk factors for non-Hodgkin's lymphoma:

- Some viruses are implicated, including:
  - Epstein-Barr virus: associated with Burkitt's lymphoma, Hodgkin's disease, lymphomas in immunocompromised patients (eg, from HIV infection, organ transplantation) and sino-nasal lymphoma.
  - Hepatitis C is associated with certain subtypes.
  - Kaposi's sarcoma-associated herpesvirus is associated with lymphomas in patients with HIV infection.

- Congenital and acquired immunodeficiency states.
- Autoimmune disorders: eg, Sjögren's syndrome and Hashimoto's thyroiditis, promote the development of MALT and predispose patients to subsequent lymphoid malignancies.
- *Helicobacter pylori* infection is associated with the development of primary gastrointestinal lymphomas, especially gastric MALT lymphomas.

Investigations:

- FBC: anemia (marrow failure or autoimmune hemolytic anemia), thrombocytopenia and neutropenia. Thrombocytosis and lymphocytosis may also occur.
- Renal function and electrolytes: obstructive nephropathy, hypercalcaemia.
- LFTs.
- Serology: HIV, HTLV-1, hepatitis C.
- Cytogenetic and immunochemistry play a significant role in the diagnosis of Burkitt's lymphoma.
- CXR: mediastinal adenopathy, pleural or pericardial effusions and parenchymal involvement.
- Biopsy: Lymph node excision or adequate core biopsy is required for the diagnosis of follicular lymphoma.
- CT scan of the neck, chest, abdomen and pelvis: to detect enlarged lymph nodes, hepatosplenomegaly. This is the most widely used investigation for initial staging and assessment of treatment response.



TOPIC: Polycythemia/Erythrocytosis

Definition:

Erythrocytosis is an abnormally high haematocrit and haemoglobin concentration.

- **Absolute erythrocytosis** occurs when there is an increase in the total number of red blood cells in the circulation (increased red-cell mass). This is classified as:
  - **Primary erythrocytosis caused by polycythaemia vera,** when the increased red-cell mass is caused by neoplastic proliferation of hematopoietic cells in the bone marrow. Since these cells are capable of developing into platelets, red blood cells, and neutrophils, polycythaemia vera may also be accompanied by thrombocytosis, neutrophil leucocytosis, and splenomegaly. Approximately 95% of people with polycythaemia vera have a V617F mutation in exon 14 of the JAK2 gene, encoding Janus kinase, a regulator of myeloproliferation. A further 4% have mutations in exon 12 of JAK2, and the remaining 1% have other mutations including LNK mutations.
  - **Secondary erythrocytosis,** when the increased red-cell mass is caused by increased production of erythropoietin. Most commonly erythropoietin is increased due to a physiological response to chronic tissue hypoxia caused by chronic lung disease, such as chronic obstructive pulmonary disease. Inappropriate production of erythropoietin by the kidney can also occur with kidney disorders (such as renal artery stenosis), benign and malignant renal tumours, and hepatomas. Other causes include sleep apnoea, obesity-related hypoventilation, testosterone therapy or performance-enhancing drug use (anabolic steroids or erythropoietin), or hereditary genetic mutations affecting the oxygen-sensing pathway or haemoglobin oxygen affinity.
  - **Idiopathic erythrocytosis,** when the increased red-cell mass has no identifiable cause. Some people have a latent cause which becomes apparent in time. In other people, the condition resolves spontaneously.
- **Apparent erythrocytosis** is defined as increased haematocrit and haemoglobin concentration with a normal red-cell mass. It is caused by a reduced plasma volume and most commonly occurs in people with a high body mass index, those taking diuretics, or people who are heavy smokers or heavy consumers of alcohol.

Complications and prognosis:

- **Erythrocytosis leads to increased blood viscosity and elevated risk of thrombosis.**
- **For people with polycythaemia vera:**
  - Overall, the risk of thrombosis is 1.6-fold higher than in the background population. Cardiovascular risk and mortality rate are increased accordingly.
  - The risk of haemorrhage may also be increased, particularly in people with severe thrombocytosis.
  - Aquagenic pruritus is a common clinical feature of polycythemia vera, and may present in 31–69% of people with the condition.
    - Polycythaemia vera associated pruritus can have a significant impact on quality of life and personal hygiene, and can lead to people abandoning bathing completely.
  - People requiring cytoreductive therapy (usually those at high risk of thrombosis) are most often managed with hydroxycarbamide. Pancytopenia is the most common serious adverse drug reaction.
    - Risk may be mitigated with appropriate monitoring.
    - **Note**: People with polycythaemia vera who develop cytopenias while receiving hydroxycarbamide treatment may be at increased risk of transformation to leukaemia and death.

- **For people with secondary erythrocytosis:**
  - The prognosis is largely determined by the nature of any underlying condition.
- **For people with apparent erythrocytosis:**
  - Evidence from small non-randomised studies indicates increased morbidity and mortality. However, it is unclear whether this is directly due to increased haematocrit values, or related to other factors.
  - There is no robust evidence that reducing the haematocrit reduces morbidity or mortality.

Diagnosis:

- **Be aware that erythrocytosis may be found incidentally as a result of a full blood count (FBC) taken from a person who has not reported symptoms.** The full blood count should be repeated after a minimum of one week to see whether the rise is transient. Previous results should also be rechecked to determine whether the trend was previously present.
- **Suspect erythrocytosis in people experiencing symptoms potentially caused by hyperviscosity, including:**
  - Chest and abdominal pain.
  - Myalgia and weakness.
  - Fatigue.
  - Headache — may be described as a sense of 'fullness' in the head and neck, with dizziness, and/or perspiration.
  - Tinnitus.
  - Blurred vision, temporary loss of vision in one or both eyes.
  - Paresthesia.
  - Slow mentation, sense of depersonalisation.
- **Additional symptoms that may indicate polycythaemia vera as the cause of erythrocytosis include:**
  - Bruising.
  - Pruritis, especially on contact with warm water.
  - Abdominal discomfort (relating to splenomegaly).
  - Insomnia.
  - Vasomotor symptoms, particularly hot flushes, accompanying hyperhidrosis or night sweats.
  - Tenderness or painful burning and/or redness of fingers, palms, heels, or toes.
    - **Note**: 10–15% of people with polycythaemia vera are diagnosed following an acute thrombotic event, and 2–8% following major haemorrhage.
- **Take a full history:**
  - **Ask about/note factors which may indicate polycythaemia vera, including:**
    - Age over 40 years.
    - Personal history of haemorrhage, thrombosis, or Budd-Chiari syndrome.
    - Family history of polycythaemia vera.
  - **Ask about/note factors which may indicate apparent erythrocytosis, including:**

        - Obesity.
        - Smoking.
        - Alcohol excess.
        - Hypertension.
        - Use of thiazide diuretics.
    - **Ask about/note factors which may underlie secondary erythrocytosis, such as:**
        - Cardiac and respiratory symptoms or disease, smoking, and potential exposure to carbon monoxide.
        - Excessive daytime sleepiness, snoring, sleep disturbances, and observed night-time episodes of apnoea (particularly in people who are obese or with other respiratory conditions).
        - Previous renal transplantation.
        - Performance enhancing drug use (testosterone, anabolic steroids or erythropoietin).
- **Perform an examination:**
    - Note whether there is a ruddy complexion (indicative of erythrocytosis).
    - Examine the eyes to assess whether conjunctival plethora (indicative of erythrocytosis) is present.
    - Palpate the abdomen to elicit:
        - Splenomegaly, which can be indicative of polycythaemia vera — however, bear in mind that a palpable spleen is present in only 40% of cases.
        - Abdominal masses. Benign and malignant uterine, renal, and hepatic tumours, which may be palpable, can secrete erythropoietin, leading to secondary erythrocytosis.
    - Examine the digits, measure oxygen saturation, and listen to the chest — clubbing of digits, oxygen saturation of less than 92% in room air, and/or abnormal heart or breath sounds may suggest secondary erythrocytosis caused by underlying cardiopulmonary disease.
    - Carry out urine dipstick analysis to identify possible renal causes of secondary erythrocytosis.
- **Consider** differential diagnosis
- **Take blood samples (without a tourniquet if possible) for haemoglobin, mean corpuscular volume (MCV), haematocrit, white blood cell count (WBCs), platelet count, liver function tests (LFTs), urea and electrolytes, and e-GFR.**
    - The British Society for Haematology recommend that people with a persistently raised haematocrit (>0.52 in males and >0.48 in females), or with a raised red cell mass (>25% above predicted) should be investigated.
    - A significant proportion of people with these findings will have apparent erythrocytosis (where the red cell mass is normal but the plasma volume is reduced).
    - The results of these tests, alongside clinical judgement (based on noted signs and symptoms and the presence/absence of risk factors for apparent erythrocytosis), should; therefore, be used to determine which people require immediate additional testing. As a guide:
        - One-off values of elevated Hct (>0.6 in males and >0.56 in females) indicate an absolute erythrocytosis and require additional investigations.

        - Polycythaemia vera is suggested by raised white blood cell (leukocytosis) and platelet counts (thrombocytosis) in addition to a high haemoglobin and haematocrit. MCV is usually low in polycythaemia vera. LFTs are usually normal, but elevated values raise the possibility of Budd-Chiari syndrome. People exhibiting findings suggestive of polycythaemia vera should have additional investigations.
        - Secondary erythrocytosis may be suggested by abnormal LFTs, if there is a hepatic tumour. Abnormal renal function can indicate secondary erythrocytosis with an underlying renal cause.
- **Additional investigations that may be carried out to confirm, refute, and refine a diagnosis include:**
    - Serum erythropoietin — increased serum erythropoietin suggests secondary erythrocytosis, while decreased levels suggest polycythaemia vera. Normal levels do not rule out polycythaemia vera.
    - Serum ferritin — to screen for iron deficiency if a low MCV is found. Can support a diagnosis of polycythaemia vera in people with suggestive signs or symptoms, but clinically normal haemoglobin.
    - JAK2 V617F mutation — where this is positive in addition to a clear clinical history and haematological features it is considered definitive of polycythaemia vera. A negative finding does not rule it out, as around 5% of cases involve other mutations.
    - Serum uric acid (optional) — frequently elevated in polycythaemia vera.
    - Abdominal ultrasound — to detect splenomegaly (suggestive of polycythaemia vera) where physical examination is equivocal.

What else might it be?

The differential diagnosis of polycythaemia vera includes:

- **Essential thrombocythaemia (ET)** — can be difficult to distinguish clinically from polycythaemia vera. Splenomegaly is infrequent, and JAK2 mutations are less common than in polycythemia vera. A normal erythropoietin, isolated thrombocytosis, and absence of elevated haemoglobin concentration are suggestive of essential thrombocytosis.
- **Chronic myelogenous leukaemia (CML)** — may be clinically indistinguishable from polycythaemia vera; although, thrombosis is much less common with CML. Diagnosis is confirmed by screening for Philadelphia chromosome 9;22 translocation or its fusion protein product: BCR-ABL.
- **Congenital polycythaemia** — an inherited condition which is present from birth, although symptoms may first develop during childhood or adulthood. Occurs due to a mutation in the genes encoding the erythropoietin receptor or proteins involved in the regulation of erythropoietin (EPOR, VHL, EGLN1, or EPAS1). Congenital polycythaemia can have different inheritance patterns depending on the affected gene.

Management:

From age 18 years onwards.
How should I manage a person with erythrocytosis/polycythaemia?

- **If polycythaemia vera is suspected, and/or the person is experiencing symptoms of hyperviscosity, refer urgently to a haematologist for treatment.**
- **Specialist advice should also be sought if there is any uncertainty regarding the diagnosis.**
- **If apparent erythrocytosis is suspected:**

- Address factors that may lead to reduced plasma volume, including smoking, alcohol consumption, obesity, and hypertension.
- If the person is taking a thiazide diuretic, consider switching to an alternative anti-hypertensive agent.
- People with confirmed apparent erythrocytosis should be offered a blood test every 3-6 months in primary care, and if their haematocrit is greater than 0.54 on two consecutive occasions, the person should be referred to haematology to commence a venesection programme.

- **If secondary erythrocytosis is suspected and thought due to the use of performance-enhancing drugs:**
  - Advise that the person avoids the use of performance-enhancing drugs such as erythropoietin, testosterone supplements, and anabolic steroids.
    - Blood sampling should be repeated two months after these measures are implemented, to determine whether the haematocrit has normalised. If not, further tests should be carried out to rule out other causes of secondary erythrocytosis and polycythaemia vera.
- **If secondary erythrocytosis is suspected and thought due to other causes, refer to an appropriate specialist to ensure optimal management of the underlying cause.**
- **For all people with polycythemia/erythrocytosis, manage cardiovascular disease risk factors,** including hyperlipidaemia, diabetes, hypertension, and smoking.
  - **Note**: iron replacement should be used judiciously in people with erythrocytosis.
- **People with polycythaemia vera require annual follow-up,** which can be done in primary care to detect the 2-8% who will transform to myelofibrosis or the 1-3% who will transform to acute myeloid leukaemia.
  - Features suggestive of myelofibrosis include:
    - Anaemia.
    - Sustained loss of requirement for either venesection (in the absence of cytoreductive therapy) or cytoreductive treatment for erythrocytosis.
    - A leukoerythroblastic peripheral blood picture.
    - Increasing splenomegaly, defined as either an increase in palpable splenomegaly of greater than, or equal to, 5 cm (distance of the tip of the spleen from the left costal margin), or the appearance of a newly palpable splenomegaly.
    - Development of at least one of three constitutional symptoms: greater than 10% weight loss in 6 months, night sweats, or unexplained fever (>37.5°C).
  - Cardiovascular risk should be assessed around the time of diagnosis and annually using a validated clinical risk assessment tool.

How is polycythaemia vera treated in secondary care?

- **The recommended management of all people with polycythaemia vera in secondary care includes:**
  - **An explanation** of the risks and implications of the condition.
  - **Venesection** to maintain the haematocrit at less than 0.45.
    - The volume of blood removed and frequency of venesection are adjusted depending on the person's size and tolerability to venesection.

        - Phlebotomy can result in iron deficiency which can induce symptoms of fatigue and impact cognition. However, iron supplementation is generally contraindicated as recurrent polycythaemia can emerge rapidly. A short trial of iron replacement with careful monitoring may be indicated for people who experience systemic symptoms of iron deficiency. This should only be undertaken by an experienced specialist in secondary care.
    - **Prescription of aspirin 75 mg daily** (unless this is contraindicated).
- **In addition, pharmacological cytoreductive therapy is recommended for people at high risk of thrombosis** (people over the age of 60 years, or those with any history of thrombosis). The first-line drug is usually hydroxycarbamide, with interferon alfa or ruxolitinib as possible alternatives where hydroxycarbamide is contraindicated, not tolerated, or ineffective. Busulfan may be used in people with an advanced age or with a short life expectancy where hydroxycarbamide, interferon alfa, or ruxolitinib have not been tolerated.
    - Treatments are typically titrated over a course of several weeks to maintain the haematocrit at less than 0.45.
    - Hydroxycarbamide resistance is associated with a higher risk of disease transformation to leukaemia and death.
- **Pharmacological cytoreductive therapy may also be considered if:**
    - White blood cell or platelet count is abnormally high.
    - Thrombosis or disease-related bleeding occurs.
    - There is evidence of disease progression, such as weight loss or night sweats.
    - Splenomegaly progresses or becomes symptomatic.
    - The need for venesection is frequent or persistent, or the person is poorly tolerant of venesection.
    - The person's symptom burden is too high, or where pruritis is severe despite treatment.
- **Alternative treatments may be offered to special populations such as pregnant women and children.**
    - Polycythaemia vera is rare in pregnancy, and haematocrit concentrations may improve due to increases in the maternal plasma volume. Pregnant women at high risk of thrombosis may be treated with interferon alfa, while those at low risk may be managed by phlebotomy and low-dose aspirin alone.
    - Asymptomatic children at low risk of thrombosis may not receive cytoreductive or antithrombotic therapy owing to the risk of side effects. Children at high risk may be cautiously treated with low-dose aspirin (owing to the risk of Reye's syndrome in under 12s). Hydroxycarbamide and interferon alfa treatment is reserved for use in rare cases where other management options have been ineffective.
- **People with polycythemia vera will be closely monitored under the direction and supervision of a haematology specialist.**
- Aquagenic pruritus is a common clinical feature of polycythemia vera.
    - Where venesection or cytoreductive therapy has failed to control the symptoms, antihistamines may be considered, but evidence of efficacy is conflicting.
    - Other adjunctive treatments may include H2-receptor antagonists (ranitidine or cimetidine — for dual-histamine blockade), tricyclic antidepressants, selective serotonin reuptake inhibitors, and antiseizure medications such as gabapentin or pregabalin.

TOPIC: Von Willebrand's disease

Von Willebrand's disease is the most common inherited bleeding disorder. The majority of cases are inherited in an autosomal dominant fashion* and characteristically behaves like a platelet disorder i.e. epistaxis and menorrhagia are common whilst haemoarthroses and muscle haematomas are rare.

Clinical features:

- **Bruising and mucocutaneous bleeding** — Bruising and mucocutaneous bleeding are common, including:
  - Easy bruising
  - Cutaneous bleeding
  - Prolonged bleeding from mucosal surfaces (eg, oropharyngeal, gastrointestinal, uterine)
- **Heavy menstrual bleeding**
- **Postpartum bleeding**

Management:

Reassure and explain

- usually, mild form
- inherited
- Due to low factor in blood that cause bleeding

Advice

- Educate patients as to the bleeding risk. Provide advice regarding drugs that must be avoided such as non-steroidal anti-inflammatory drugs and antiplatelet drugs.
- First aids if bleeding

Prescription:

- tranexamic acid for mild bleeding
- desmopressin (DDAVP): raises levels of vWF by inducing release of vWF from Weibel-Palade bodies in endothelial cells
- factor VIII concentrate

Referal:

- Refer to haematology to confirm diagnosis

Investigation:

- prolonged bleeding time
- APTT may be prolonged
- factor VIII levels may be moderately reduced
- defective platelet aggregation with ristocetin
- CBC and coagulation profile usually normal.
- Thrombocytopenia Individuals with type 2B VWD may have mild thrombocytopenia
- Screening tests for VWD include VWF antigen (VWF:Ag), platelet-dependent VWF activity (VWF:RCo or VWF:GPIbM), and factor VIII activity

Prevention:

- After the diagnosis of vWD has been made, testing should be offered to first-degree relatives (with or without a positive bleeding history).

TOPIC: Haemophilia

Hemophilia A (factor VIII [8] deficiency) and hemophilia B (factor IX [9] deficiency) are heritable X-linked recessive bleeding disorders. Severity depends largely on the factor activity level

Symptoms:

- Haemoarthroses:
  - Hemarthrosis is the most common site for bleeding in ambulatory patients
- haematomas
  - Common sites of bleeding include joints and soft tissues, muscles, oral mucosa, and intracranial hemorrhage (ICH).
- prolonged bleeding after surgery or trauma

Management :

Prescription:

- Hemophilia A: replace coagulation factor VIII
- Hemophilia B: replace coagulation factor IX
  - Over time antibodies can develop to the replacement coagulation factors.
- Desmopressin (DDAVP) can be used in acute episodes of mild haemophilia A to promote the function of von Willebrand factor and only if bleeding persists would coagulation factor concentrates be used.
- In patients with severe disease, coagulation factors should be given prophylactically and self-administered through intravenous access.
- Other mainstays of treatment include:
  - Analgesia
    - Paracetamol
    - Opioids
    - Avoid the use of aspirin and NSAIDs due to their increased risk of bleeding.
  - Rest, ice, compress and elevate for musculocutaneous injuries.
  - Physiotherapist involvement
  - In severe haemorrhage, antifibrinolytic agents can be considered.
  - **Lifestyle changes such as avoiding contact sports**

Referral:

To hematology

Investigation:

- CBC
- prolonged APTT
- bleeding time, thrombin time, prothrombin time normal
- Factor IX percentage activity.
- Factor VIII mutation analysis.
- Abdominal ultrasound and endoscopy may be necessary if gastrointestinal bleeding is a feature.

TOPIC: Cryoglobulinaemia

Immunoglobulins which undergo reversible precipitation at 4 deg C, dissolve when warmed to 37 deg C. One-third of cases are idiopathic

Three types:

- type I (25%):
  - monoclonal – IgG or IgM
  - associations: multiple myeloma, Waldenstrom macroglobulinaemia
- type II (25%)
  - mixed monoclonal and polyclonal: usually with rheumatoid factor
  - associations: hepatitis C, rheumatoid arthritis, Sjogren's, lymphoma
- type III (50%)
  - polyclonal: usually with rheumatoid factor
  - associations: rheumatoid arthritis, Sjogren's

Possible features

- Raynaud's only seen in type I
- cutaneous
  - vascular purpura
  - distal ulceration
  - ulceration
- arthralgia
- renal involvement
  - diffuse glomerulonephritis

Investigations

- low complement (esp. C4)
- high ESR

Management:

- treatment of underlying condition e.g. hepatitis C
- immunosuppression
- plasmapheresis

TOPIC:Tranexamic acid

Tranexamic acid is a synthetic derivative of lysine. Its primary mode of action is as an antifibrinolytic that reversibly binds to lysine receptor sites on plasminogen or plasmin. This prevents plasmin from binding to and degrading fibrin.

Tranexamic acid is most commonly prescribed to help treat menorrhagia.

The role of tranexamic acid in trauma was investigated in the CRASH 2 trial and has been shown to be of benefit in bleeding trauma when administered in the first 3 hours. Tranexamic acid is given as an IV bolus followed by an infusion in cases of major haemorrhage.

There is also ongoing research looking at the role of tranexamic acid in traumatic brain injury.

TOPIC: MGUS

Monoclonal gammopathy of undetermined significance (MGUS, also known as benign paraproteinaemia and monoclonal gammopathy) is a common condition that causes a paraproteinaemia and is often mistaken for myeloma. Differentiating features are listed below. Around 10% of patients eventually develop myeloma at 10 years, with 50% at 15 years

Features:

- usually, asymptomatic
- no bone pain or increased risk of infections
- around 10-30% of patients have a demyelinating neuropathy

Differentiating features from myeloma:

- normal immune function
- normal beta-2 microglobulin levels
- lower level of paraproteinaemia than myeloma (e.g. < 30g/l IgG, or < 20g/l IgA)
- stable level of paraproteinaemia
- no clinical features of myeloma (e.g. lytic lesions on x-rays or renal disease)

TOPIC: Neutropenic sepsis

Neutropenic sepsis is a relatively common complication of cancer therapy, usually as a consequence of chemotherapy. It most commonly occurs 7-14 days after chemotherapy. It may be defined as a neutrophil count of < 0.5 * 10^9 in a patient who is having anticancer treatment and has one of the following:

- a temperature higher than 38ºC or
- other signs or symptoms consistent with clinically significant sepsis

Prophylaxis:

- if it is anticipated that patients are likely to have a neutrophil count of < 0.5 * 10^9 as a consequence of their treatment they should be offered a fluoroquinolone

Management:

- antibiotics must be started immediately, do not wait for the WBC
- NICE recommends starting empirical antibiotic therapy with piperacillin with tazobactam (Tazocin) immediately
- many units add vancomycin if the patient has central venous access but NICE do not support this approach
- following this initial treatment patients are usually assessed by a specialist and risk-stratified to see if they may be able to have outpatient treatment
- if patients are still febrile and unwell after 48 hours an alternative antibiotic such as meropenem is often prescribed +/- vancomycin
- if patients are not responding after 4-6 days the Christie guidelines suggest ordering investigations for fungal infections (e.g. HRCT), rather than just starting therapy antifungal therapy blindly
- there may be a role for G-CSF in selected patients

TOPIC: Paroxysmal nocturnal haemoglobinuria

Paroxysmal nocturnal haemoglobinuria (PNH) is an acquired disorder leading to haemolysis (mainly intravascular) of haematological cells. It is thought to be caused by increased sensitivity of cell membranes to complement (see below) due to a lack of glycoprotein glycosyl-phosphatidylinositol (GPI). Patients are more prone to venous thrombosis

Pathophysiology:

- GPI can be thought of as an anchor which attaches surface proteins to the cell membrane
- complement-regulating surface proteins, e.g. decay-accelerating factor (DAF), are not properly bound to the cell membrane due a lack of GPI
- thrombosis is thought to be caused by a lack of CD59 on platelet membranes predisposing to platelet aggregation

Features:

- haemolytic anaemia
- red blood cells, white blood cells, platelets or stem cells may be affected therefore pancytopaenia may be present
- haemoglobinuria: classically dark-coloured urine in the morning (although has been shown to occur throughout the day)
- thrombosis e.g. Budd-Chiari syndrome
- aplastic anaemia may develop in some patients

Diagnosis:

- flow cytometry of blood to detect low levels of CD59 and CD55 has now replaced Ham's test as the gold standard investigation in PNH
- Ham's test: acid-induced haemolysis (normal red cells would not)

Management:

- blood product replacement
- anticoagulation
- eculizumab, a monoclonal antibody directed against terminal protein C5, is currently being trialled and is showing promise in reducing intravascular haemolysis
- stem cell transplantation

TOPIC: Splenomegaly

Massive splenomegaly

- myelofibrosis, chronic myeloid leukemia, visceral leishmaniasis, malaria

Other causes (as above plus)

- portal hypertension e.g. secondary to cirrhosis
- lymphoproliferative disease e.g. CLL, Hodgkin's
- haemolytic anaemia, hepatitis, glandular fever, infective endocarditis, RA

Major causes of splenomegaly

Congestive
Cirrhosis
Heart failure
Thrombosis of portal, hepatic, or splenic veins
Malignancy
Lymphoma, usually indolent variants
Acute and chronic leukemias
Polycythemia vera
Multiple myeloma and its variants
Essential thrombocythemia
Primary myelofibrosis
Primary splenic tumors
Metastatic solid tumors
Infection
Viral – Hepatitis, infectious mononucleosis, cytomegalovirus
Bacterial – *Salmonella*, *Brucella*, tuberculosis

Parasitic – Malaria, schistosomiasis, toxoplasmosis, leishmaniasis
Infective endocarditis
Fungal
Inflammation
Sarcoid
Serum sickness
Systemic lupus erythematosus
Rheumatoid arthritis (Felty syndrome)
Infiltrative, nonmalignant
Gaucher disease
Niemann-Pick disease
Amyloid
Other lysosomal storage diseases (eg, mucopolysaccharidoses)
Langerhans cell histiocytosis
Hemophagocytic lymphohistiocytosis
Rosai-Dorfman disease
Hematologic (hypersplenic) states
Acute and chronic hemolytic anemias, all etiologies
Sickle cell disease (children)
Following use of recombinant human granulocyte colony-stimulating factor

TOPIC: Waldenstrom's macroglobulinaemia

Waldenstrom's macroglobulinaemia is an uncommon condition seen in older men. It is a lymphoplasmacytoid malignancy characterised by the secretion of a monoclonal IgM paraprotein

Features:

- systemic upset: weight loss, lethargy
- hyperviscosity syndrome e.g. visual disturbance
    - the pentameric configuration of IgM increases serum viscosity
- hepatosplenomegaly
- lymphadenopathy
- cryoglobulinaemia e.g. Raynaud's

Investigations:

- monoclonal IgM paraproteinaemia
- bone marrow biopsy is diagnostic
    - infiltration of the bone marrow with lymphoplasmacytoid lymphoma cells

Management:

- typically, rituximab-based combination chemotherapy

TOPIC: Wiskott-Aldrich syndrome

Wiskott-Aldrich syndrome causes primary immunodeficiency due to a combined B- and T-cell dysfunction. It is inherited in a X-linked recessive fashion and is thought to be caused by mutation in the WASP gene.

Features:

- recurrent bacterial infections (e.g. Chest)
- eczema
- thrombocytopaenia
- low IgM levels